

The Standard Model Symmetry Atlas and Exhaustion Closure

Target-indexed normal forms, physical role,
and same-scope exhaustion

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SYM-11 of the Standard Model Symmetry Arc

Abstract

The word “symmetry” in the Standard Model covers structures that differ in their acting objects, physical quotients, anomaly status, vacuum realization, global form, operator locus, and renormalization-group scope. This paper gives a target-indexed capstone classification for a declared family of Standard Model symmetry targets. The classified object is a typed physical statement rather than a group name. After lawful redescription, every admitted same-target first difference is routed through nine formation loci and 54 source-theorem subfamilies. For each admitted target/subfamily pair, exact source hypotheses define a target-qualified normal-form class space; named atlas rows are optional representatives and their population is not used as a completeness argument.

QFT status is fixed before structural role. A memberwise class-side physical projection is constructed independently of role burdens, residues, atlas occupancy, and compatibility tests. It is then matched, by exact physical-signature equality, to an independently derived source-branch range. This yields a full-domain role-occupancy theorem with assignment-level compatibility transport, including the already proved early-sector occupancy orbits. Carrier support is derived from source-licensed operations and proved removal-minimal relative to the frozen direct-support presentation. A disjoint rival theorem then partitions any proposal into lawful equivalence, same-target obstruction, target update, or genuine same-target counterexample; the generated genuine branch is empty.

Consequently, every closure-eligible target fiber has a unique atlas up to lawful redescription and no unclassified same-scope symmetry-relevant survivor. The theorem is relative to the declared target family, record language, source hypotheses, role architecture, and support criteria. It does not assert uniqueness of the Standard Model under arbitrary BSM or UV premises, select a unique global form, or determine numerical parameters. Full atlas tables, audit-grade theorem statements, source-first derivations, and reproducibility materials accompany the paper but are not premises of the scientific proof.

Principal theorem

Fix a closure-eligible target $D \in \mathfrak{D}_{\text{SM}}^{\text{cl}}$. Let $\mathfrak{A}_D$ be the canonical typed symmetry atlas, let $\equiv_D$ be lawful identity-preserving redescription, and let C_D be the independently generated same-target candidate universe. For any complete admissible atlas $\mathfrak{A}'_D$ satisfying the same target, record language, candidate protocol, source-theorem hypotheses, role architecture, and support criteria,

$$\mathfrak{A}'_D / \equiv_D \cong \mathfrak{A}_D / \equiv_D, \quad \text{Surv}_D^{\text{sym}}(C_D \setminus \mathfrak{A}_D) = \emptyset.$$

The isomorphism preserves target-qualified physical/status identity, role incidence, support ancestry, and source/evidence status; it need not preserve names, notation, row order, or other descriptive skin. The result is same-scope and target-indexed, not an unrestricted uniqueness theorem for all extensions of the Standard Model.

Results and reading guide

SYM-11 is an integration theorem rather than another sector catalogue. The upstream papers separately classify gauge, spacetime, anomaly, discrete, flavor, chiral/custodial, accidental, electroweak-vacuum, global-form/generalized, and scale/RG symmetry structures. The present paper proves that those sector results compose into one target-indexed normal form and that no additional same-target symmetry-relevant structure survives the declared candidate and comparison protocol.

The four scientific burdens

1. **Candidate completeness.** A same-target physical difference must enter an independently defined formation locus and a source-theorem subfamily; named-row population cannot define the domain.
2. **Redescription-stable physical identity.** Gauge, basis, normalization, coordinate, and scheme changes are quotiented only when every dependent physical/status field is transported together.
3. **Compatible role and support.** Physical residues and theorem burdens are constructed independently; pointwise eligibility is not enough, and full assignment compatibility is transported to every theorem-defined specialized class. Carrier support is derived from explicit source-licensed operations and tested by provider recomputation after deletion.
4. **Rival elimination.** A proposal is equivalent, same-target obstructed, target-changing, or a genuine same-target counterexample. The genuine branch remains open until the source-first classification and composite calculus discharge it.

Proof architecture

The theorem identifiers encode dependency order rather than importance:

Layer	Scientific task	Main output
F01–F03	Freeze target family, imports, and typed record substrate	Non-circular comparison domain
Q01–Q04	Normalize records and lawful redescription	Canonical extensional quotient $\mathfrak{A}_D/\equiv_D$
Q05–Q06	Generate and classify candidates	Target-qualified normal-form class spaces $\mathcal{N}_{D,\ell,\alpha}$
Q07–Q10	Factor authority and prove compatible occupancy	One lawful full-domain role orbit
Q11–Q13	Prove support completeness/minimality and construction coherence	Source-licensed relative-minimal support graph
Q14–Q15	Partition and eliminate rivals	Empty generated genuine same-target branch
A01–A04	Assemble fibers and target updates	Atlas uniqueness, same-scope exhaustion, and family closure

Scientific proof spine

1. Freeze the target graph, generation-governance handoff, target updates, record schema, lawful equivalence, and import firewall.
2. Route every same-target physical/status first difference through one of nine formation spaces, then through one of 54 typed source-theorem subfamilies.
3. Define the target-qualified source normal-form class spaces before any atlas population, role, residue, support provider, or rival conclusion is read.
4. Construct theorem burdens and physical residues independently; evaluate complete pointwise

eligibility and assignment-level compatibility.

5. Lift occupancy to the theorem-defined specialized classes by an independently computed class-side physical signature, exact source-branch equality, and compatibility-preserving transport.
6. Derive carrier-neutral support requirements and source-licensed providers, then prove retained edges indispensable under provider recomputation.
7. Apply the disjoint rival theorem and discharge the genuine same-target branch from candidate completeness and finite composite closure.
8. Assemble the fiberwise atlas isomorphisms and same-scope exhaustion theorem over the licensed target-update graph.

Companion architecture

This publication manuscript contains the complete proof at journal granularity. The accompanying audit-grade theorem volume preserves every target/subfamily bridge, generated theorem table, and bounded hostile regression [1, Secs. 1–18 and Apps. B–K]. The atlas companion exposes the canonical rows and synchronized target, role, support, rival, and provenance views [2, companion title page and synchronized atlas views]. The source-first derivation report documents the reconstruction boundary, the 54 source bridges, the class-first Q10 transport, support removal, endpoint derivation, and reproducibility boundary [3, Secs. 1–3, especially Secs. 2.2–2.7]. These artifacts certify the implementation of the theorem; they do not replace its semantic proof.

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Theorem identity across the release layers

The publication manuscript uses compact theorem names, while the audit-grade volume preserves stable internal identities and the source-first report exposes the corresponding reconstruction surfaces [1, 3]. The table is an identity crosswalk, not a new dependency layer.

Table 1. Publication-to-audit theorem identity and source-first reconstruction map

Publication theorem or proof layer	Audit-grade identity and location	Source-first report location
F01–F03; Q01–Q04	Same compact IDs; audit-grade Secs. 2–5	Sec. 1 reconstruction boundary; Sec. 2.1 source-first transfer theorem
Q05 candidate-locus completeness	SYM11-Q05; audit-grade Sec. 6.1	Secs. 2.1–2.2, transfer theorem and 54 source bridges
Q06 generated-candidate status completeness	SYM11-Q06; audit-grade Secs. 6.2–6.4	Secs. 2.2–2.3, 54 bridges and frozen-domain/NF realization
Q06 class-projection range and source-branch totality	SYM11-V210-Q06-CLASS-PROJECTION-RANGE-AND-SOURCE-BRANCH-TOTALITY; audit-grade Sec. 8.4	Sec. 2.5, provenance-independent equivalence and exact occupancy transport
Independent class-first full-domain Q10 lift; Q09/Q10	SYM11-V210-SPECIALIZED-NF-INDEPENDENT-PI-CLASS-BETA-P-PUP-J-TRANSPORT ; SYM11-Q09/Q10; audit-grade Secs. 8.4–8.6	Secs. 2.4–2.5, materialized occupancy and class-first $P/P_{\text{up}}/J$ transport
Q11–Q12 support completeness and removal minimality	SYM11-Q11; SYM11-Q12; audit-grade Sec. 9	Sec. 2.6, source-licensed support and relative semantic removal
Q14–Q15 rival partition and genuine-rival elimination	SYM11-Q14; SYM11-Q15; audit-grade Secs. 12–13	Sec. 2.7 endpoint derivation
A01–A04; SYM11-MAIN	SYM11-A01–SYM11-A04; SYM11-MAIN; audit-grade Secs. 14–15	Sec. 2.7 endpoint derivation

Citation rule

When a compressed proof step depends on a theorem-specific bridge or finite source surface, the publication manuscript cites the audit-grade theorem location and the exact source-first section. The atlas companion is cited only as an inspectable downstream projection, never as completeness or uniqueness authority.

I

Scope, target family, and proof authority

1 Why a symmetry atlas needs a closure theorem

A conventional summary of Standard Model symmetry often starts with the local gauge group, adds Poincaré invariance, lists accidental global charges, and then appends comments about anomalies and electroweak breaking. That shorthand is useful pedagogically, but it is not a closure object. It suppresses distinctions that become decisive as soon as one asks whether two claims are the same physical structure, whether a symmetry acts on the physical quotient, whether it survives quantization or vacuum selection, or whether it exists only in a limit or enlarged operator category. The underlying distinctions are standard across constrained gauge systems, spacetime representation theory, anomaly analysis, flavor invariants, generalized symmetries, and RG/EFT descriptions [4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20].

The completed sector arc makes the failure of a flat list explicit. SYM-1 separates gauge-product, transition, transport, representation, reduction, boundary, topology, BRST, and anomaly data rather than assigning one status to “gauge symmetry” [21]. SYM-2 separates the Minkowski/Poincaré domain anchor, active transformations, passive coordinates, Noether diagnostics, particle representations, and the CPT theorem premise package [22]. SYM-3 shows that gauge inconsistency, anomalous current nonconservation, ’t Hooft obstruction, global anomaly, inflow, and matching are different quantum statuses [23]. SYM-4–SYM-7 then show that discrete, flavor, chiral, custodial, and accidental symmetries are sector-, parameter-, vacuum-, operator-, and neutrino-target indexed [24, 25, 26, 27]. SYM-8–SYM-10 make the target problem unavoidable by adding electroweak fixation, global form and defects, and RG/EFT scope [28, 29, 30].

1.1 A list of names is not a set of theorem objects

A candidate symmetry claim requires at least the following data:

1. the target theory, field content, phase, boundary/topology, global form, operator category, quantum regime, and scale interval;
2. the acting mathematical object and the object acted upon;
3. the action or transformation law;
4. the invariant or covariant physical content;
5. the physical quotient;
6. current, constraint, Ward, cohomological, or defect data;
7. anomaly status;
8. vacuum or state realization;
9. explicit, anomalous, topological, vacuum, or scale-dependent breaking ancestry;
10. structural role and evidence authority.

The same familiar name can point to different records because any one of those fields changes. Compact-support local gauge transformations and asymptotic gauge transformations share a Lie algebra parameter but differ in physical quotient and charge. A flavor basis transformation and a zero-Yukawa flavor action both act on generation indices but have different invariants and quotient status. Classical scale invariance and RG covariance both use scale language but act on different objects.

1.2 Inventory, normalization, and exhaustion

Three tasks must be kept separate.

Inventory. Identify candidate records and populate the fields.

Normalization. Split composite names, merge genuine aliases, and quotient presentation-equivalent rows.

Exhaustion. Prove that every admitted same-target proposal enters the candidate grammar and that no generated genuine rival survives.

A complete inventory does not automatically prove normalized uniqueness. A normalized atlas does not automatically prove candidate completeness. An empty rival ledger does not prove exhaustion. The capstone therefore places the closure burden in explicit theorems rather than in a database state.

The failure mode the capstone is designed to block

The invalid chain is

$$\text{many rows} \implies \text{complete inventory} \implies \text{unique classification} \implies \text{exhaustion.}$$

None of the implications is automatic. Completeness requires a generator theorem; uniqueness requires a lawful equivalence and compatible occupancy theorem; exhaustion requires a rival theorem with the genuine branch scientifically discharged.

1.3 The two-track architecture

The capstone preserves the arc's two-track order [31]:

$$\text{QFT status and physical quotient} \longrightarrow \text{typed atlas record} \longrightarrow \text{AASC role}$$

Track A determines what the symmetry claim means in ordinary field-theoretic language. Track B asks what standing-bearing work the already typed structure performs. A role label never serves as the proof that a transformation is gauge, physical, anomalous, exact, emergent, or broken.

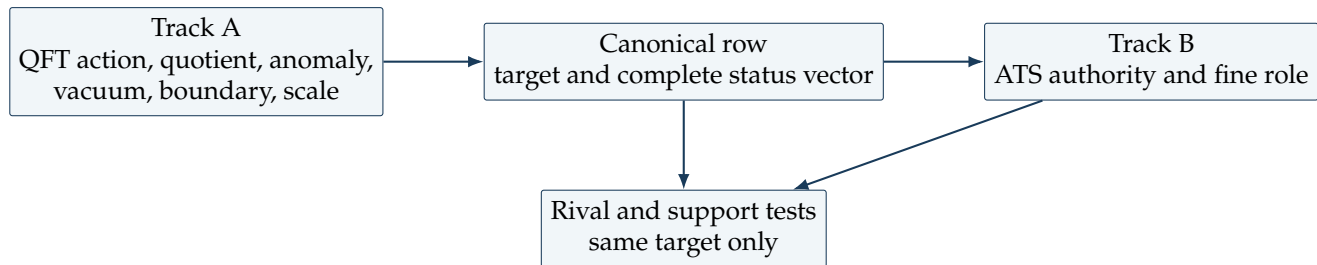


Figure 1: The capstone preserves QFT status as logically prior to role assignment and rival disposition.

1.4 The capstone novelty

The main novelty is not the existence of a table. It is the proof that the table is a target-indexed normal form. The result has four layers:

1. a theoremic candidate-locus completeness result;
2. a redescription-stable row quotient;
3. unique compatible role occupancy and removal-minimal carrier support;

4. a same-scope rival theorem and achieved relative exhaustion.

The atlas companion is useful because it makes the result inspectable. The theorem does not derive its authority from the companion's row count.

2 Frozen target family, generation governance, and comparison discipline

The capstone target is a graph, not a universal theory and not the Cartesian product of every scope coordinate appearing anywhere in the arc. A node is admitted only when a frozen sector theorem supplies a scientifically coherent target. An edge records a licensed update and the exact claims that survive or fail across it.

Definition 2.1 (Target-family graph). Let

$$\mathfrak{D}_{\text{SM}}^{\text{sym}} = (V_D, E_D)$$

be the finite declared graph whose nodes are fully typed Standard Model symmetry targets and whose arrows carry source, destination, changed coordinates, preserved interfaces, invalidated claims, and source theorem. Let $\mathfrak{D}_{\text{SM}}^{\text{cl}} \subset V_D$ be the closure-eligible nodes: those for which all sector-specific status, quotient, anomaly, vacuum, operator, and scale burdens used by SYM-11 are frozen.

A node carries at least field content, generation governance where applicable, operator order, phase/state, faithful global-form status, operator category, line-polarization branch where applicable, monopole policy, boundary/topology, background action or fractionalization data, quantum regime, scale/EFT interval, spin scope, backgrounds, and neutrino convention. Those coordinates are not decorative metadata. They determine whether the acting object exists, whether a quotient is defined, whether an anomaly statement is meaningful, and whether a line, limit, or defect structure is part of the target.

2.1 Generation-cardinality reception

Generation cardinality is not re-proved in the Symmetry ARC and is not entered as a symmetry row. The Carrier ARC proves in the open-generation arena that

$$\text{ClosedGenCert}(n) \iff n = 3,$$

and its downstream handoff permits the fixed Standard Model target to be instantiated at $N_g = 3$ without adding a primitive generation carrier [32].

Proposition 2.2 (Generation-target reception). *The core Standard Model symmetry target receives $N_g = 3$ as a target-governance datum. The datum is inherited by Standard Model fibers unless the target explicitly changes generation structure. It is neither an additional symmetry action nor a new primitive carrier. A proposal with $N_g \neq 3$ is therefore a generation-target update or a direct challenge to the upstream cardinality theorem, not a missing symmetry-atlas row.*

Proof. The imported Carrier-ARC result has theorem authority only for the role-occupancy endpoint and its handoff. SYM-11 uses the conclusion solely to fix the Standard Model target before flavor, mixing, anomaly, and remnant records are populated. No step of the SYM-11 proof infers $N_g = 3$ from the dimension of CKM/PMNS matrices, observed family names, empirical non-observation of a fourth generation, or atlas row count. $\square$

Target fingerprint

The base node $D_{\text{SM}}^{\text{core}}$ is the minimal renormalizable Standard Model on Minkowski spacetime with the chiral matter pattern at $N_g = 3$, one Higgs doublet, anomaly-neutral local gauge architecture, selected electroweak phase, minimal no- ν_R convention, and the sharp-local point-particle/operator category. Controlled fibers include boundary and topological gauge targets, BRST presentations, neutrino variants, SMEFT truncations, the three-flavor chiral target, separate exact and physically deformed isospin and hadronic- $SU(3)$ targets, scalar/gauge-Higgs/minimal-matter/Dirac-matter custodial targets, target-qualified heavy-spin/heavy-flavor/combined-spin-flavor/matched-HQET fibers, fixed global forms with explicitly fixed line branches and monopole policy, massless defect targets, fixed-point and EFT envelopes, and the restricted homogeneous vacuum sector.

2.2 Neutrino-target separation and SYM-7 source targets

The target freeze receives the SYM-7 neutrino and operator strata without collapsing physically distinct theories. In particular, the renormalizable Majorana target

$$D_{\text{NURM}} = \{\text{SM} + 3\nu_R, M_R \neq 0, d \leq 4\}$$

is distinct from the low-energy Weinberg target

$$D_{\text{NUW}} = \{\text{minimal SM without dynamical } \nu_R, \kappa \neq 0, d \leq 5\}.$$

Their field content, breaking tensor, operator order, and basis/phase data differ. A heavy-Majorana matching map may connect them, but it is an inter-target EFT relation rather than representative equivalence. Accordingly, D-SM-NU-MAJORANA denotes the renormalizable three- ν_R branch, while D-SMEFT-D5 denotes the Weinberg branch. Existing low-energy Majorana CP and PMNS records remain on the latter target.

The capstone also publishes a source-target crosswalk for all thirteen SYM-7 registered schemas. The minimal, electroweak-topological, Dirac- ν_R , renormalizable Majorana, Weinberg, $d = 6$ SMEFT, $d = 6$ ν SMEFT, $d = 7$, selected $d = 9$, and two $B - L$ background targets are represented explicitly. The parameterized remnant schema is instantiated only by its twelve frozen effect sets. The registered Dirac- ν_R plus electroweak-topology successor schema is retained explicitly outside $\mathfrak{D}_{\text{SM}}^{\text{cl}}$, because the frozen thirty-four-result SYM-7 ledger exports no independent canonical result on that combined target. This is a visible theorem boundary, not silent target collapse.

Theorem 2.3 (Complete target-family and governance freeze, SYM11-F01). *The declared data define a target graph $\mathfrak{D}_{\text{SM}}^{\text{sym}}$ with a well-defined closure-eligible subfamily $\mathfrak{D}_{\text{SM}}^{\text{cl}}$. For every admitted comparison of two symmetry records, exactly one of the following holds:*

1. *their complete target fingerprints agree, so they enter the same-fiber comparison relation; or*
2. *at least one target-bearing coordinate differs, and the comparison is mediated by a registered update arrow or lies outside the declared family.*

For Standard Model fibers the fingerprint includes the received $N_g = 3$ governance datum. For fixed- Γ line targets it additionally includes the global quotient, operator category, maximal mutually local line-polarization branch, monopole policy, theta branch, relevant background-action/fractionalization data, and spin scope. No closure theorem in this manuscript quantifies over an unregistered Cartesian product of target coordinates.

Proof. Each node in the target registry contains every mandatory coordinate and a closure status. Each update arrow records a nonempty changed-coordinate set and distinct source and destination IDs. Equality of target fingerprints is ordinary fieldwise equality on the target schema. The generation-governance registry fixes $N_g = 3$ before Standard Model row comparison. The four SYM-9 global-completion nodes specify the line branch and monopole policy explicitly rather than by an unspecified

phrase such as “declared branch.” If all target fields agree, the comparison is same-fiber. If any target field differs, the records are not objects of one comparison fiber. When the destination is registered, the corresponding source theorem supplies the update arrow; otherwise the proposal is outside $\mathfrak{D}_{\text{SM}}^{\text{sym}}$. These cases are mutually exclusive and exhaustive because the target schema is frozen before row comparison. The no-Cartesian claim follows because only registry nodes, not arbitrary coordinate tuples, are members of V_D . $\square$

2.3 No silent Cartesian completion

The global-form family illustrates the point. SYM-9 admits four compact connected central quotients, but a fixed- Γ line target also carries operator category, line polarization, monopole policy, background action, and physical phase data [29]. It does not follow that every such fiber is automatically crossed with every neutrino variant, every thermal phase, and every EFT truncation. A cross-product node must be scientifically licensed.

The same restraint applies to the massless non-invertible targets, the zero-Yukawa higher-flavor target, the custodial limits, and the homogeneous vacuum sector. They are controlled fibers, not generic properties of the physical core target.

2.4 Same-target comparison and target update

Definition 2.4 (Same-target relation). For records s and s' , write $s \sim_{\text{tgt}} s'$ when their target IDs and every target-bearing coordinate agree. Write $s \rightsquigarrow s'$ when an admitted target-update arrow carries the first target to the second.

Theorem 2.5 (Target-fiber separation, SYM11-Q02). *For all admitted records s, s' , same-target comparison and target update are disjoint. Every admitted target-bearing difference is represented by an update arrow; every same-target rival preserves the complete target fingerprint. A candidate cannot be rescued from failed same-target status by silently changing generation cardinality, field content, operator order, phase, global form, line branch, monopole policy, operator category, boundary/topology, quantum regime, scale/EFT envelope, backgrounds, or neutrino convention.*

Proof. Disjointness follows from the definitions: $s \sim_{\text{tgt}} s'$ requires equality of every target field, while an update arrow has a nonempty changed-coordinate set. Exhaustiveness over admitted changes follows from SYM11-F01. The final statement is the fixed-target rule: once a candidate has been submitted against a target D , its disposition is evaluated in the fiber over D . Altering a target field creates a new comparison, not a repair of the old one. $\square$

The complete target, generation-governance, and update registries are supplied in the audit-grade theorem volume [1, Sec. 2 and App. C] and the atlas companion [2, “Target-family nodes and licensed updates].

3 Proof authority, imports, and the non-circularity firewall

SYM-11 is a composition theorem, but it is not allowed to import its conclusion. The sector papers supply the physics of the objects they own. The capstone-support manuscripts supply fixed-domain governance interfaces. Neither source class is treated as a substitute for the locally proved candidate-completeness, occupancy, rival, and atlas-uniqueness arguments.

3.1 Proof-authority hierarchy

The proof distinguishes four authorities.

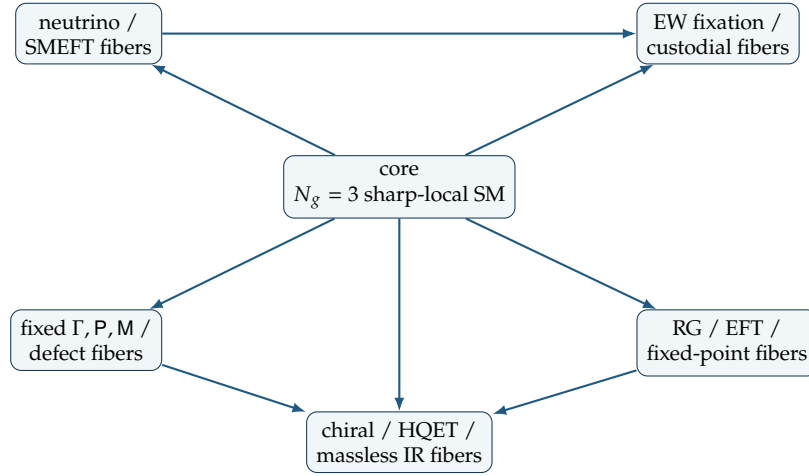


Figure 2: Schematic target-family graph. The target family contains only licensed nodes and updates; arrows are not an invitation to form every Cartesian product.

Authority	Permitted burden	Prohibited promotion
Primary QFT and sector theorem	Transformation law, quotient, anomaly, vacuum, operator, boundary, and scale status	A citation does not establish capstone completeness by repetition
Local capstone proof	Target graph, normalization, locus completeness, occupancy, support, rival partition, atlas isomorphism	A registry label does not replace the proof
Restricted AASC support	Fixed-domain ATS decomposition, role-occupancy logic, same-scope closure form, trajectory coherence, audit standing	A prior global or Standard-Model uniqueness conclusion is not imported
Artifact audit	Schema, identifiers, referential integrity, selected encoded relations, provenance, synchronization, reproducibility	The scripts do not infer physics or prove semantic exhaustion

The sector theorem chain is *physics-first*: SYM-1 classifies gauge loci, SYM-2 spacetime action, SYM-3 anomalies, SYM-4 discrete symmetries, SYM-5 flavor and basis, SYM-6 chiral / custodial / IR limits, SYM-7 accidental symmetries, SYM-8 electroweak fixation, SYM-9 global form and generalized symmetry, and SYM-10 scale / RG / emergence [21, 22, 23, 24, 25, 26, 27, 28, 29, 30]. SYM-0 supplies the common typed record and role vocabulary [33].

3.2 Restricted support imports

The ten capstone-support manuscripts are used only through the following interfaces.

- Fixed-domain exhaustion supplies the form of a theoremic locus reduction and the rule that a boundary import changes scope rather than repairing a failed same-scope construction [34].
- Anchor/Tensor/Skin supplies the fixed-domain authority trichotomy, not the fine symmetry taxonomy [35].
- Role-occupancy closure supplies the distinction among cardinality, pointwise eligibility, live residue, and joint compatible realization [36].
- Same-scope operator closure supplies no-deferred-admissibility and the distinction between lawful

redescription and strict enlargement [37].

- Coherent construction supplies path, prefix, and composition discipline for versioned atlas construction [38].
- Claim standing supplies evidence ancestry and reportability discipline; it is governance support rather than a truth oracle [39].
- The remaining fixed-domain and Standard-Model closure manuscripts supply terminology for standing quotient, fixed-load comparison, and non-circular forcing. Their terminal uniqueness conclusions are not premises here [40, 41, 42, 43].

Imported interface

The import registry records the exact use and blocked inference of every source. In particular, no entry permits the inference “an earlier AASC capstone is unique, therefore the symmetry atlas is unique.” The present atlas theorem must be reconstructed from SYM-0–SYM-10, the frozen target family, and the local capstone lemmas.

Theorem 3.1 (Import firewall, SYM11-F02). *No imported proposition in the dependency graph has the same antecedent and conclusion as SYM11-MAIN. More strongly, after deletion of every prior terminal uniqueness or exhaustion conclusion, all premises used below remain available: sector status theorems, fixed-domain typing, ATS authority decomposition, role-occupancy definitions, same-scope enlargement rules, and construction coherence. Hence the capstone conclusion is not an imported premise or a restatement of one.*

Proof. The conclusion of SYM11-MAIN quantifies over the target graph $\mathfrak{D}_{\text{SM}}^{\text{sym}}$, the canonical thirty-field record classes, the nine same-fiber candidate loci, the target-qualified sector-export crosswalk, the independently constructed role-demand and eligibility structures, the carrier-support relation with removal witnesses, and the disjoint rival disposition. None of the restricted support manuscripts contains these SYM-11 objects. Conversely, every support conclusion retained in the import registry is lower-level: it constrains fixed-domain authority, admissible occupancy, enlargement, or construction paths without identifying the members of $\mathfrak{A}_D$. The sector papers identify those members but do not prove the family-level candidate-locus theorem or fiberwise atlas isomorphism. Removing imported terminal closure claims therefore leaves the premises of every local proof intact. The dependency graph is acyclic because no SYM-0–SYM-10 theorem imports SYM-11, and no support import is allowed to cite the present atlas as evidence. $\square$

Adversarial check

A circularity objection has standing only if it identifies a specific imported proposition whose conclusion already contains the SYM-11 candidate universe, role occupancy, support graph, rival elimination, or atlas-isomorphism result. Similar vocabulary or an earlier fixed-domain closure theorem is not enough.

The complete source and import maps are supplied in the audit-grade theorem volume [1, Sec. 3 and App. B].

4 Canonical symmetry record and atlas schema

A symmetry name is not the classified object. The classified object is a target-indexed statement with an acting object, an acted-upon object, a transformation law, a physical quotient, observable content, quantum and vacuum status, operator and scale scope, evidence, and role. The atlas therefore preserves enough structure to prevent one familiar word from carrying several incompatible claims.

Definition 4.1 (Canonical symmetry record). For $D \in \mathfrak{D}_{\text{SM}}^{\text{sym}}$, a record is

$$s_D = (D, G, X, \rho, q, O, \Omega, \mathcal{A}, \mathcal{V}, \mathcal{B}, \mathcal{R}, \sigma, \mathcal{E}),$$

where G is the acting object, X the acted-upon object, ρ the action, q the physical quotient, O the invariant or covariant physical content, Ω the current/constraint/charge data, $\mathcal{A}$ the anomaly data, $\mathcal{V}$ the vacuum or state realization, $\mathcal{B}$ the boundary/global-form/operator-background data, $\mathcal{R}$ the scale/EFT range, σ the structural-role package, and $\mathcal{E}$ the evidence, rival, and nonclaim package.

The compressed Track-A status vector is

$$\text{SymStat}_D(s) = (Q, C, Q_m, V, B, G_{\text{op}}, R, A, E),$$

with quotient action, classical status, quantum status, vacuum realization, boundary/global sensitivity, generalized-operator locus, scale/EFT range, anomaly type, and epistemic status. Structural role is assigned only after this vector is fixed.

4.1 Thirty-field publication row

The public row expands the record into the thirty fields frozen by the ARC plan: immutable identity and aliases; target and background; candidate, acting object, acted-upon object and action mode; invariant object and quotient; current/constraint/charge; classical, quantum, anomaly, vacuum, breaking, and scale fields; operator and global-form fields; role and role proof; carrier support; evidence and source; rival status; nonclaims; paper owner; and change history. Six internal fields record target ID, candidate locus, authority channel, role instance, theorem IDs, and implementation status.

A field can be NA only with a typed reason. “Unknown,” “not checked,” and “not applicable” are different states. An open scientific burden remains visible and prevents closure of the corresponding target fiber.

Theorem 4.2 (Atlas-schema freeze, SYM11-F03). *The frozen schema defines a stable record datatype, mandatory fields, allowed candidate loci, authority channels, stable theorem identifiers, evidence statuses, and supersession relations. Any change that alters the physical or mathematical content of a row creates a new row version and a supersession edge; a change of name, layout, or other certified skin may retain the record identity.*

Proof. The schema separates the thirty canonical publication fields from the six internal control fields. Every field has a declared semantic role. Record identity is extensional in the target and physical/status fields, not in row order or display name. Therefore a content-preserving edit is a new presentation of one object, whereas a content-changing edit changes the extensional signature and must receive a new version. The supersession relation records ancestry without erasing the prior statement. $\square$

Theorem 4.3 (Atlas well-formedness, SYM11-Q01). *Every canonical atlas row is a well-formed instance of the frozen schema: each mandatory field is populated or carries a reasoned not-applicable value; every target, locus, role instance, carrier, source, theorem, rival, and supersession reference resolves; and every row’s Track-A status precedes its AASC role conclusion.*

Proof. The proof has a scientific and an implementation component. Scientifically, each row is the normalized export of an owning SYM sector theorem, which fixes its target, action, quotient, anomaly, vacuum, boundary/operator, and scale status before role assignment. Cross-owned rows are reconciled by the order in [section 10](#). Formally, the schema map sends each record component to its publication field and rejects missing mandatory coordinates. Referential integrity then follows from the frozen target, theorem, source, role, carrier, and supersession registries. This establishes well-formedness of the canonical rows. It does not yet prove that the rows are complete; that is the burden of [section 6](#). $\square$

Physics translation

A complete row is not a claim that every aspect of the underlying physics has been solved. It is a claim that the status being published is fully scoped and that any unresolved physical burden is represented explicitly rather than hidden by the word “symmetry.”

The canonical atlas summary is:

Canonical atlas release summary

The canonical atlas contains **270** record classes over **78** declared target nodes and **62** licensed update arrows. The independent capstone proof layer freezes **9** candidate formation spaces, **9** locus evaluator theorems, **531** theorem-domain/locus proof cells, **10** independent sector role modules, and **16167** carrier-neutral support requirements. Source-population crosswalks and the atlas are downstream projections of those scientific constructions, not proof authority.

II**Normal forms and candidate completeness****5 Target-fiber separation and record normalization**

Before completeness can be tested, duplicate names and composite labels must be normalized. The rule is extensional: split when physical/status data differ; merge only when every standing-bearing field agrees and a lawful redescription carries all dependent data together.

Definition 5.1 (Extensional identity signature). For a record s in target D , define

$$\text{Id}_D(s) = (D, \text{Act}, \text{Obj}, q, O, \text{SymStat}_D(s), \text{OpLocus}, \text{ATS}, \text{Role}).$$

Aliases, row order, gauge coordinates, basis coordinates, normalization choices, and equivalent scheme representatives are excluded unless a sector theorem shows that they alter one of the displayed entries.

Definition 5.2 (Lawful record equivalence). Write $s \equiv_D s'$ when s and s' have the same target and an admitted redescription transports their acting and acted-upon objects, quotient, observables, anomaly, vacuum, boundary/global-form, operator, scale/EFT, and role data into one another. The transport must include every dependent field; partial coordinate changes do not define equivalence.

Theorem 5.3 (Redescription stability, SYM11-Q03). *The relation $\equiv_D$ is an equivalence relation, and the extensional identity signature is constant on its classes. Consequently the Track-A status and fine role descend to $\mathfrak{A}_D/\equiv_D$.*

Proof. Reflexivity is the identity presentation. Symmetry uses the inverse of the admitted redescription; every admitted gauge, basis, normalization, chart, or scheme map in the sector papers is invertible on the comparison class to which it is applied. Transitivity uses coherent composition with all dependent interfaces transported. Each defining field of Id_D is invariant or covariant under the corresponding sector theorem: proper gauge operations preserve the quotient class; passive spacetime and flavor-basis changes preserve physical actions and invariants; rephasings preserve invariant CP data; hypercharge and charge-unit normalizations preserve the transported charge/global packet; gauge and field-coordinate choices preserve the electroweak packet; and scheme changes preserve quotient RG content [21, 22, 24, 25, 28, 29, 30]. Hence the signature and role are class functions. $\square$

5.1 Split, merge, alias, and parent–child rules

The normalization rewrite system has four operations.

1. **Split**: one familiar label becomes several rows when target, quotient, anomaly, vacuum, operator, boundary, or scale status differs.
2. **Merge**: two rows become one only when their extensional signatures agree and a lawful redescription is exhibited.
3. **Alias**: a second name points to the same canonical class without creating a new record.
4. **Parent–child**: a target–fiber specialization remains a distinct row linked by a target–update edge.

Familiar label	Required split	What may merge
Gauge symmetry	architecture, proper-local operation, boundary action, large/topological action, BRST	gauge choices inside one quotient locus
CP	strong, weak/Yukawa, Higgs, neutrino, theta and target-specific sectors	rephasing-equivalent presentations with equal invariants
Flavor symmetry	zero-Yukawa action, basis covariance, spurion bookkeeping, fixed-tensor stabilizer	weak-basis representatives of one invariant packet
Baryon/lepton number	perturbative, anomalous/nonperturbative, neutrino, SMEFT and remnant fibers	normalization-equivalent charge descriptions
Custodial symmetry	scalar, gauge–Higgs, matter and neutrino-qualified targets	invertible bidoublet presentations
Electroweak breaking	gauge representative, fixation class, residual action, pole/spectrum and Goldstone descriptions	gauge-fixed pictures of one physical phase
One-form symmetry	electric, magnetic, global-form, screening and monopole-policy fibers	line-coordinate presentations preserving the genuine spectrum
Scale symmetry	dilatation, RG transport, scheme change, fixed point, EFT matching and emergence	scheme representatives of one quotient RG trajectory

Theorem 5.4 (Canonical record normalization, SYM11-Q04). *Repeated application of the split, merge, alias, and parent–child rules terminates in a canonical row normal form. Two normalized rows in one target fiber are identical exactly when their extensional signatures agree; target–fiber specializations remain linked but distinct.*

Proof. Let the normalization measure be the ordered tuple consisting of unresolved composite-status collisions, duplicate extensional signatures, untyped aliases, and target differences represented as same-fiber rows. A split reduces the first coordinate; a merge reduces the second; aliasing reduces the third; and parent–child conversion reduces the fourth. No operation increases an earlier coordinate, so termination follows by well-founded lexicographic descent. Local confluence follows because each rule is decided by the same extensional signature and target relation: splits are forced by unequal fields, merges by equality plus lawful transport, aliases by equality without independent row content, and parent–child relations by target inequality. Thus every reduction path reaches the same normalized classes. $\square$

Adversarial check

The phrase “same group” or “same transformation matrix” is not a merge certificate. The target, quotient, state, operator, anomaly, boundary, scale, and physical-action fields must agree. Conversely, different notation is not evidence for two physical records.

6 Candidate generation and total scientific disposition

The capstone uses four scientific layers in the strict order

schema formation $\longrightarrow$ typed subfamily selection
 $\longrightarrow$ proved local classification
 $\longrightarrow$ canonical atlas projection.

No later object may define an earlier one. Source-row population, canonical record IDs, target closure labels, selected role occupants, support providers and rival-ledger state therefore have no authority in candidate formation or classification.

6.1 First-difference completeness and the nine formation spaces

Let $\mathcal{R}_D$ be the frozen F03 record language at target D . Its physical and status fields route to

$$\mathcal{L}_D = \{\text{ACT}, \text{QUO}, \text{ANOM}, \text{VAC}, \text{BND}, \text{OPR}, \text{PAR}, \text{DIM}, \text{RG}\}.$$

Target-bearing differences exit the same-fiber comparison; presentation-only differences enter the lawful-redescription quotient; identifiers and certification fields create no scientific candidate.

Lemma 6.1 (Complete first-difference routing). *If $s, s' \in \mathcal{R}_D$ have the same target, are not lawfully presentation-equivalent, and differ in physical or status content, then their first physical difference lies in one of the nine loci above. A field not represented by the published first-difference map falsifies the lemma openly.*

Proof. The complete F03 field set is partitioned into target-bearing, physical/status, presentation and downstream metadata fields. Action, acting object, acted-on object, invariant and current data route to ACT; representative and physical-quotient transport to QUO; anomaly and quantum obstruction to ANOM; state, stabilizer and fixation to VAC; global-form, boundary, topology and line data to BND; local/extended operator and defect data to OPR; exact parameter strata and deformation data to PAR; operator-order and complete grammar data to DIM; and RG, matching, fixed-point and emergence data to RG. The complete field-to-locus map and its source-first construction are supplied in the audit-grade theorem volume [1, Sec. 6.1 and App. D.1] and source-first report [3, Secs. 2.1–2.2]; an omitted physical/status field would falsify the routing theorem rather than being repaired by relabeling. $\square$

For each $\ell \in \mathcal{L}_D$, let $F_{D,\ell}$ be the corresponding target-qualified formation space. These spaces are defined from target coordinates and typed variables alone; their nine constructors are stated in the machine-readable technical supplement.

Theorem 6.2 (Independent candidate-locus completeness, SYM11-Q05). *For every admitted symmetry-relevant proposal s' at frozen target D , exactly one of the following occurs:*

1. *a target coordinate changes, hence the proposal is a target update or outside the theorem domain;*
2. *all physical/status data agree after lawful dependent transport, hence s' belongs to an existing $\equiv_D$ orbit; or*
3. *the first physical/status difference determines $\ell \in \mathcal{L}_D$ and an element of $F_{D,\ell}$.*

Consequently

$$C_D = \bigcup_{\ell \in \mathcal{L}_D} F_{D,\ell}$$

relative to the frozen record language. An unrouteable same-target physical difference is a counterexample to Q05.

Proof. Apply Lemma 6.1. Formation uses the changed typed datum and the frozen target, never an existing atlas row. The theorem domain is the independently frozen $59 \times 9 = 531$ target-locus surface and is invariant under mutations of target closure labels or sector-export population. $\square$

6.2 The second partition and the 54 actual classification theorems

Locus membership is not yet scientific classification. For every locus the formation space is partitioned into typed subfamilies

$$F_{D,\ell} = \bigsqcup_{\alpha \in A_{D,\ell}} F_{D,\ell,\alpha}.$$

Across the nine loci there are 54 subfamilies. The selector α is determined from typed acting object, action, quotient, anomaly, state, operator, parameter or continuation data before theorem lookup and before any atlas projection.

Proposition 6.3 (Selector partition). *For every $x \in F_{D,\ell}$ there exists exactly one selector tag α such that $x \in F_{D,\ell,\alpha}$. A typed candidate matching no tag falsifies exhaustiveness; overlapping tags falsify disjointness. There is no default “known row” branch.*

Proof. The selectors are constructors of the F03 scientific datatype. Their mutually exclusive object/action/status/operator predicates and their exhaustive constructor coverage are frozen independently of source-row population in the frozen subgenerator registry. The complete $59 \times 54 = 3186$ target–subfamily surface is then generated from these selectors. Removing a sector-export row does not alter the formation or selector objects. $\square$

Lemma 6.4 (Frozen Q05 domain lock). *Let $G_\alpha(D)$ be the frozen Q05 target guard of subfamily $F_{D,\ell,\alpha}$. If $G_\alpha(D) = 1$, then (D, ℓ, α) is a Q06 classification burden independently of whether a canonical NF representative exists at D . In particular,*

$$G_\alpha(D) = 1 \not\Rightarrow \text{NF}_{D,\ell,\alpha} \neq \emptyset,$$

and

$$\text{NF}_{D,\ell,\alpha} = \emptyset \not\Rightarrow \text{TGT}.$$

The TGT branch is available only when the candidate changes at least one frozen target coordinate.

Proof. The Q05 guard is fixed before theorem lookup and before atlas projection. All 1,909 admitted target–subfamily pairs are evaluated on this frozen surface. For every admitted pair the exact source equations define a target-qualified normal-form class space modulo Q03. A named atlas row is optional downstream materialization and its absence has no dispositional force. A same-target failed source equation is OBS; an actual change of field content, operator order/category, global form, background, phase/state, scale/EFT or another frozen target coordinate is TGT; anything not decided by the cited source partition is OPEN. The implementation audit checks equality of the frozen Q05 domain and the Q06 burden surface, but the target-specialization theorem supplies the scientific classification. $\square$

A decisive dependency rule for Q06 is that the Q05 target/subfamily guard is the authoritative classification domain. A selector no longer points merely to a label stating “total”, Q04 normalization alone is never treated as proof of canonical-atlas membership, and absence of a canonical NF realization never removes a Q05-admitted target from the Q06 burden. Every subfamily has an explicit local proposition

$$L_{\ell,\alpha} : F_{D,\ell,\alpha} \longrightarrow \{\text{NF}, \text{OBS}, \text{TGT}, \text{OPEN}\}$$

with an exact source-packet projection, proof that its kernel is precisely lawful redescription, exact source hypotheses and authority, a total branch classifier, and a theorem-defined target-specialized source normal-form class space (or a proved no-new-orbit/frozen-registry case). All 54 theorem bodies appear in the audit-grade theorem volume [1, Sec. 6.2 and App. K.2]; the same source-domain, projection-faithfulness, remainder, and NF-realization burdens are reconstructed in the source-first report [3, Sec. 2.2]. The machine-readable theorem registry is their projection rather than their proof authority.

Lemma 6.5 (Projection–faithfulness–classification lift). *Let F be a capstone candidate subfamily, S an exact upstream theorem packet, and*

$$\pi : F \longrightarrow S$$

a total source-packet projection. Assume:

1. π preserves every target and identity-bearing physical/status coordinate used by the capstone comparison;
2. $\ker \pi$ is exactly the already proved lawful-redescription relation;
3. the source theorem partitions S exclusively and exhaustively into normal-form, obstruction, target-update and explicit-open branches; and
4. every source normal-form solution has a target-qualified lawful class in $\mathcal{N}_{D,\ell,\alpha}$, with named Q04 rows only optional representatives.

Then F has an exclusive and exhaustive NF/OBS/TGT/OPEN disposition, and every NF element belongs to exactly one lawful capstone orbit represented in $\mathfrak{A}_D/\equiv_D$.

Proof. Totality of π sends every $x \in F$ to the source theorem domain. The source partition supplies exactly one branch. Preservation of target-bearing data prevents a changed-target source branch from being relabeled as same-target NF; preservation of obstruction data prevents an obstruction from being erased; and explicit OPEN is transported unchanged. In the NF branch, two capstone elements with the same source normal form differ by an element of $\ker \pi$, hence are lawfully equivalent. Conversely, the target-specialization theorem places every source-normal-form solution in its own lawful class in $\mathcal{N}_{D,\ell,\alpha}$. It does not identify distinct classes. Q04 may supply a named materialized representative when one exists, but row materialization is downstream. Thus an additional satisfying orbit is retained as an element of the canonical normal-form class space rather than disappearing outside the atlas. Failure of any hypothesis makes the local theorem OPEN rather than allowing a totality label to stand in for the missing implication. $\square$

For each of the 54 subfamilies, the corresponding bridge theorem in the audit-grade theorem volume verifies these four hypotheses at the exact cited source scope. An upstream finite-inventory or normal-form theorem is used only after a capstone bridge proves that the selected formation domain maps faithfully into that finite packet; otherwise the subfamily remains OPEN.

Target specialization is *target and subfamily specific* and strictly downstream of the frozen Q05 domain. Across the frozen closure family, the Q05 guards admit 1,909 target–subfamily pairs; every one remains a Q06 burden irrespective of named-row availability. For each admitted pair the source-domain predicate and source normal-form equations define $\mathcal{N}_{D,\ell,\alpha}$ modulo Q03. Named records are optional representatives whose complete normalization authority must lie inside the bridge dependency closure. If a same-target candidate satisfies the exact source normal-form equations, its Q03 class belongs to $\mathcal{N}_{D,\ell,\alpha}$ even when no named record exists. Failure of a mandatory same-target source equation is OBS; TGT requires an actual change of a frozen target coordinate; unresolved source authority remains OPEN. The explicit HQET gauge specialization is witnessed by the HQET covariant-derivative and field-strength facts together with the SYM-1 gauge transport/connection facts. The authoritative target-specialization space registry and disposition surface are supplied in the audit-grade theorem volume [1, Sec. 6.2 and Apps. D.1, K.3] and reconstructed source-first in Sec. 2.3 of the derivation report [3, Sec. 2.3]; a source-satisfying candidate outside the corresponding class space or an unresolved theorem interface reopens Q06/Q15.

Lemma 6.6 (Local classification totality). *For every registered $F_{D,\ell,\alpha}$ and every x in that subfamily, exactly one branch holds:*

$$\text{NF}, \quad \text{OBS}, \quad \text{TGT}, \quad \text{OPEN}.$$

Furthermore,

$$L_{\ell,\alpha}(x) = \text{NF} \implies [x]_{\equiv_D} \in \mathfrak{A}_D/\equiv_D.$$

Thus equation satisfaction without orbit classification is not sufficient for NF.

Proof. For each subfamily, the audit-grade theorem volume states the exact source hypotheses and conclusion, constructs the map from capstone formation variables into the source packet, proves that the map retains every identity-bearing field and has only lawful-redescription kernel, and then applies the source theorem at that exact scope. Lemma 6.4 prevents the target domain from being redefined by the support of the NF map. The target-specialization certificate identifies the complete retained physical/status equations of the source NF branch and defines their Q03 quotient $\mathcal{N}_{D,\ell,\alpha}$ at the unchanged target. Named canonical record identities are optional representatives of elements already in this quotient and are never used to decide whether the NF branch exists. A same-target source-satisfying candidate outside this quotient, or a missing source theorem needed to define it, remains OPEN and blocks Q06. Distinct satisfying Q03 classes remain distinct elements of the class space. The stated local proof also identifies the same-target obstruction and changed-target branches. If the exact theorem authority is absent or the classification does not cover the satisfying orbit space, the branch is OPEN. Hence no metadata label can turn an unresolved classification into a total one. $\square$

6.3 Finite elimination of the conservative catch-all branches

Two deliberately conservative families require special treatment: registered operator categories not otherwise selected and registered dimension/operator grammars not otherwise selected. They are not closed by an “owning theorem exists” convention. Instead the frozen registries are explicitly exhausted:

$$\begin{aligned}\mathcal{R}_{\text{OPR}} &= \bigsqcup_i \mathcal{R}_{\text{OPR},i}, \\ \mathcal{R}_{\text{DIM}} &= \bigsqcup_j \mathcal{R}_{\text{DIM},j},\end{aligned}$$

where every registered element is mapped to a named typed family. The residual “other registered” sets are therefore empty at the frozen target family. A newly registered category or grammar reopens Q06 rather than being silently absorbed. The complete finite maps are given in the audit-grade theorem volume [1, Sec. 6.3 and Apps. D.2, K.3].

Theorem 6.7 (Generated-candidate status completeness, SYM11-Q06). *For every $D \in \mathfrak{D}_{\text{SM}}^{\text{cl}}$, $\ell \in \mathcal{L}_D$, and $x \in F_{D,\ell}$, there is a unique subfamily α and a unique scientific disposition*

$$L_{\ell,\alpha}(x) \in \{\text{NF}, \text{OBS}, \text{TGT}, \text{OPEN}\}.$$

The complete physical/status disposition is fixed before record normalization and before role assignment. OPEN cannot be converted into closure by target status, source coverage, an empty rival ledger, or implementation checks.

Proof. Proposition 6.3 gives unique α ; Lemma 6.6 gives its unique scientific disposition. The two catch-all branches are closed only by the explicit finite registry theorems above. The 3186 target-subfamily surface is evaluated against the frozen guards, and every one of the 1909 Q05-admitted pairs inherits its scientific disposition from the projection-faithfulness-classification-orbit theorem chain, never from NF-map support or stored TOTAL fields. Lemma 6.4 supplies the domain lock. Any missing local theorem, source-satisfying candidate not represented in the target-specialized Q03 class space, or newly unclassified registry element creates OPEN and blocks Q15. $\square$

6.4 Cross-locus composites

A proposal may contain dependent data in more than one locus. The first-difference locus remains primary and the remaining loci become compatibility checks. The complete unordered locus-pair matrix has

$$\binom{9+1}{2} = 45$$

entries. Each pair is classified by already proved action, quotient, anomaly, vacuum, operator and continuation interfaces. A new standing-bearing datum changes target; a failed shared relation is OBS; an unresolved component is OPEN; complete compatible data remain in the composed NF orbit.

Proposition 6.8 (Composite closure). *Every composite of registered candidate subfamilies reduces to the same NF/OBS/TGT/OPEN calculus. A composite requiring an unregistered standing-conferring datum changes target or falsifies the proposition.*

Proof. Apply the complete 45-pair cross-locus constructor supplied in the audit-grade theorem volume [1, Sec. 6.4 and App. D.3]. The restricted same-scope operator-closure support is used only to show that lawful composition acts through already typed interfaces; it is not used as candidate-completeness authority. $\square$

Adversarial check

Q05 fails if a physical field lacks a route or a same-target typed proposal falls outside the nine formation spaces. Q06 fails if a selector overlaps or misses a candidate, if a Q05-admitted target/subfamily pair is removed or relabeled because no named NF row exists, if TGT is assigned without an actual target-coordinate change, if one of the 54 local propositions is absent or used outside its exact hypotheses, if a claimed NF record is not semantically the same typed candidate family, if a source-satisfying candidate is omitted from the target-specialized NF class space, if a newly registered operator/dimension family is unmapped, or if one composite pair is missing. Any OPEN branch blocks Q15. Row counts, source coverage and implementation checks have no authority to repair one of these failures.

III

Physical role, occupancy, and carrier support

7 Authority channels and fine SYM roles

The atlas uses two role levels. The first asks where fixed-target descriptive authority resides. The second asks what function the already typed structure performs. Confusing these levels would either collapse all physical actions into “Tensor” without explanation or inflate every useful role into a new primitive authority channel.

Definition 7.1 (ATS authority channel). For an admissibility-bearing contribution s at fixed target D ,

$$ATS_D(s) \in \{\text{Anchor}, \text{Tensor}, \text{Skin}\}.$$

Anchor carries target or standing-enabling architecture; Tensor carries invariant physical relation, action, response, or obstruction; Skin carries representative variation after target and physical content are fixed.

Target update is not a fourth channel. It is an arrow between target fibers.

Theorem 7.2 (ATS authority decomposition, SYM11-Q07). *Every fixed-target admissibility-bearing atlas contribution factors through Anchor, Tensor, or Skin. No fourth same-target authority channel is required by any SYM-0–SYM-10 record. A proposal that changes field content, background, global form, operator category, phase, scale/EFT envelope, or another target-bearing coordinate is represented by a target-update arrow rather than a fourth channel.*

Proof. Apply the fixed-domain ATS decomposition only after the target has been frozen [35]. If removal of the contribution makes the target or its standing-bearing architecture ill-typed, it is Anchor. If it preserves the target but changes an invariant physical action, relation, obstruction, response, or witness on the quotient, it is Tensor. If it changes only a representative after those contents are fixed, it is Skin. These tests are mutually exclusive at one locus. The complete atlas supplies examples in every sector and no row requires a fourth result. A target-bearing change fails the fixed-domain antecedent and is therefore handled by E_D , not by extending ATS. $\square$

The fine functional roles are

$$R_{\text{SYM}} = \{\text{domain anchor, carrier constraint, bridge/fixation, physical action, derived witness, skin, effective/emergent route, obstruction}\}.$$

Multi-role records are permitted only when the roles refer to distinct levels or loci. For example, Poincaré structure is a domain anchor and its active transformations are physical actions; a compact-support gauge operation is skin while gauge architecture is carrier-structural; an anomaly can be an obstruction and a derived witness.

Fine role	ATS channel	Operational condition
Domain anchor	Anchor	Changing it changes the target or the well-typing of the theorem object
Carrier constraint	Anchor	Removal destroys admissible architecture before a representative is chosen
Bridge/fixation	Anchor or Tensor	Mediates a phase/response interface; channel depends on whether it grounds architecture or records realized response
Physical action	Tensor	Changes physical states, observables, charges, defects, or selection rules
Derived witness	Tensor	Invariant, current, anomaly, degeneracy, or theorem records established structure
Skin	Skin	Changes only presentation at the same quotient and locus
Effective/emergent route	Tensor in its target	Acts physically in a stated limit, EFT, phase, or scale window without UV promotion
Obstruction	Anchor or Tensor	Invalidates target admissibility or supplies an invariant inconsistency/matching witness

Theorem 7.3 (Fine-role compatibility, SYM11-Q08). *The fine SYM role vocabulary is a functional refinement of ATS. Every role assignment has an allowed channel at its declared locus; incompatible combinations such as same-locus Skin plus unqualified physical action are excluded. Multi-role assignments are consistent exactly when their roles are attached to distinct levels/loci or are explicitly compatible within one channel.*

Proof. The operational criteria in the table determine the channel of each fine role. Apply them to the completed Track-A status, not to the row name. The sector papers establish the relevant quotient and action facts: gauge architecture/operation, active/passive spacetime action, anomaly type, basis/physical flavor split, effective-limit action, accidental-symmetry order, electroweak fixation, generalized-operator action, and RG transport. The canonical rows record their loci. A same-locus contradiction would require one operation both to change and not change physical quotient content; no canonical row has that signature. When two roles occur, their locus or logical level field witnesses the distinction. $\square$

Exact nonclaim boundary

ATS is not a replacement for the symmetry-status calculus. It classifies fixed-domain authority. The fine role vocabulary continues to carry the physics of action, witness, obstruction, bridge, and

effective route.

8 Source-first role occupancy and cross-sector compatibility

Archival identity of the compact publication headings

The publication headings below suppress two long internal identifiers for readability. Their exact audit-grade identities are SYM11-V210-Q06-CLASS-PROJECTION-RANGE-AND-SOURCE-BRANCH-TOTALITY and SYM11-V210-SPECIALIZED-NF-INDEPENDENT-PI-CLASS-BETA-P-PUP-J-TRANSPORT, both in Sec. 8.4–8.5 of the audit-grade theorem volume [1, Sec. 8.4–8.5]. Their complete source-conclusion equivalence, branch-range, named-class, P/P_{up} , and assignment-level J realizations are reconstructed in Sec. 2.5 of the source-first report [3, Sec. 2.5]. Table 1 records the wider publication-to-audit identity map.

The role theorem is not inferred from the number of atlas rows or from a previously selected assignment. Its inputs are two separately reconstructed source objects:

source hypotheses and denial consequences $\longrightarrow R_{D,p}$, source physical conclusions $\longrightarrow X_{D,p}$.

The role builder cannot inspect physical residues, atlas rows, eligibility, or occupancy. The residue builder cannot inspect roles, required interfaces, eligibility, or occupancy.

8.1 Theorem-burden role formation

For each target/module pair (D, p) , a role burden is generated by holding the target and every other theorem premise fixed and deleting one required interface. If deletion destroys a necessary theorem consequence without changing target, the lost consequence defines a role instance

$$r = (D, p, \text{required function, authority channel, physical interface}).$$

The complete burden registry contains 130 roles. Each record cites the exact source definitions/hypotheses, the denial operation, and the consequence lost. No source result object or eventual occupant appears in the constructor.

8.2 Independent physical residue extraction

Separately, the source conclusion extractor reads the actual action, quotient, invariant, anomaly, vacuum, boundary, operator, parameter, or RG statement proved by the source theorem and creates

$$x = (D, p, \text{physical proposition, typed status packet}).$$

The complete residue registry contains 130 records. The extraction process is blind to the role vocabulary and selected assignment.

Proposition 8.1 (No-backflow formation). *Changing a physical source conclusion can change $X_{D,p}$ and the subsequent eligibility relation but cannot change $R_{D,p}$. Changing a theorem burden can change $R_{D,p}$ and eligibility but cannot change $X_{D,p}$. Neither constructor reads a prior occupancy or atlas surface.*

8.3 Complete pointwise eligibility and assignment compatibility

Define

$$E_{D,p}(x, r) = 1$$

iff the independently extracted physical proposition has the target, proposition kind, action/quotient/status family, and necessary physical features demanded by the independently formed role. The source/result identifier and the semantic interface label are not used as a shortcut in the pointwise test. Every same-target role/residue pair is evaluated, giving 834 decisions.

For a complete injective assignment $\sigma : R_{D,p} \hookrightarrow X_{D,p}$, define

$$J_{D,p}(\sigma) = 1$$

iff every assigned edge is pointwise eligible and the complete physical assignment satisfies the independently frozen target, quotient, anomaly, vacuum, boundary, operator, continuation, and failure contracts. No equality $|R_{D,p}| = |X_{D,p}|$ is assumed: surplus residues remain in $X_{D,p}$ and are classified as nonlive when they occur in no J -compatible complete assignment. The construction enumerates 47 complete injective assignments before J is applied. Live residues are derived only afterward.

$J_{D,p}$ is executable rather than descriptive. Each target/module contract freezes a source-closed physical-proposition payload and the compatibility-bearing namespace (anomaly, boundary, vacuum, quotient, continuation, failure, global-form, operator, and scale status where applicable). An undeclared compatibility-bearing field is not ignored: it is an unlicensed scientific assertion and fails J . For every contract-licensed field, the declared typed relation is evaluated across the complete assignment. Thus a contradictory anomaly or continuation value cannot survive merely because the pointwise role/interface string remains unchanged.

Let $\Gamma_{D,p}^{\text{lawful}}$ act on complete assignments while preserving target identity, role incidence, the assigned residues' physical signatures, physical quotient, standing-bearing tensor data, and every source-interface certificate. The orbit invariant includes the role-to-physical-residue assignment, not merely the role vocabulary. The 42 target/module certificates satisfy

$$\left| \Sigma_{D,p} / \Gamma_{D,p}^{\text{lawful}} \right| = 1.$$

Early sectors import this exact conclusion only where the upstream paper already contains a complete occupant-independent role/eligibility/assignment construction; later sectors use capstone-local source-first constructions.

8.4 Independent class-side physical projection before branch assignment

The materialized role theorem is a proved lemma, not the definition of the final scientific domain. Let

$$X_D^{\text{full}} = A_D / \equiv_D = X_D^{\text{mat}} \cup X_D^{\text{spec}},$$

where X_D^{spec} is the union of the target-specialized Q06 normal-form class spaces $\mathcal{N}_{D,\ell,\alpha}$. The full-domain Q10 lift does not begin by assigning a Q03 class to a previously declared source branch. Instead it first constructs a physical signature from the Q06-retained coordinates of the class itself.

Definition 8.2 (Independent class-side physical projection). For every $[x] \in \mathcal{N}_{D,\ell,\alpha}$, define

$$\pi^{\text{class}}([x]) = (D, \kappa, F, I, S, V),$$

where D is the frozen target, κ the physical proposition kind, F the physical family, I the exact physical interface, S the complete physical-feature packet retained by Q06, and V the typed physical value, including every quotient, anomaly, vacuum, boundary/global-form, operator, scale, continuation, and failure coordinate inspected by the relevant occupancy theorem. The map is evaluated directly from the Q06 source-normal-form coordinates. It may not read a source-branch identifier or digest, role, residue, pointwise E , assignment-level J , atlas annotation, liveness result, materialized orbit, or endpoint status.

Source provenance remains a different object. For a source statement or conclusion packet s , write

$$\text{Src}(s) = (p, \{f_i\}, \{h_i\}, \delta_s),$$

where p is the source-owner module, f_i are exact source facts, h_i their frozen hashes, and δ_s is any certified theorem/result or theorem/theorem derivation relation. Source-owner labels, citation identifiers, theorem/table labels, hashes, and source-fact lists are therefore not coordinates of π^{class} .

Independently, the exhaustive Q06 source clauses determine a source-conclusion branch basis

$$\mathcal{C}_{D,\ell,\alpha}^{\text{src}} = \{c : c \text{ is a complete Q06-retained physical conclusion alternative at } (D, \ell, \alpha)\}.$$

For each branch c define, using the same physical schema,

$$\pi^{\text{branch}}(c) = (D, \kappa, F, I, S, V).$$

The frozen physical-interface contracts fix the possible values of this tuple before branch identifiers, roles, residues, E, J , named population, or endpoint status are read. Thus π^{class} and π^{branch} are independently formed objects with a common codomain.

Proposition 8.3 (Q03 invariance of the class-side map). *If $x \sim_{\text{Q03}} x'$ in the same target-specialized Q06 space, then*

$$\pi^{\text{class}}([x]) = \pi^{\text{class}}([x']).$$

Hence the map is well defined on Q03 classes rather than chosen representatives.

Proof. Q03 removes only the admitted presentation coordinates listed in the frozen normal-form contract. Q06 retains the target, physical proposition kind, family, exact physical interface, physical features, and typed status/value coordinates used by π^{class} . An admitted Q03 change therefore alters none of the coordinates of the class-side tuple. $\square$

8.4.1 Branch-to-space source availability is not memberwise factorization

Literal equality of citation identifiers is neither necessary nor sufficient for physical identity. Let

$$\mathcal{E}_{D,\ell,\alpha} \subseteq \mathcal{C}_D^{\text{src}} \times \{\mathcal{N}_{D,\ell,\alpha}\}$$

record only that an explicit source certificate proves a source branch to be available in the Q06 source partition of that NF space. A certificate records branch facts, NF-space facts, theorem-to-result derivation facts when needed, and the target/subgenerator locus. It may not assign a particular class member to the branch.

There are 325 such branch-to-space source-availability relations and therefore 325 specialized branch incidences. All 254 source branches receive an explicit locus disposition; 18 are proved to belong to a different Q06 locus rather than being silently treated as absent. These relations define the set of branches that may be compared with a class-side signature. They do not define π^{class} and do not choose β .

Definition 8.4 (Equality-induced memberwise branch map). For $[x] \in \mathcal{N}_{D,\ell,\alpha}$, define

$$\beta([x]) = c \iff (c, \mathcal{N}_{D,\ell,\alpha}) \in \mathcal{E} \text{ and } \pi^{\text{class}}([x]) = \pi^{\text{branch}}(c),$$

provided there is exactly one such c . Thus branch assignment is an output of exact physical-signature equality after the class-side signature has already been determined.

For an available source branch c , the exact physical fiber is consequently

$$F_{\mathcal{N},c} = \{[x] \in \mathcal{N} : \pi^{\text{class}}([x]) = \pi^{\text{branch}}(c)\}.$$

The existence of $(c, \mathcal{N}) \in \mathcal{E}$ proves only that this physical conclusion is licensed in the Q06 source domain. It cannot by itself place a member in $F_{\mathcal{N},c}$.

The dimension-five regression remains explicit. At target D-SMEFT-D5, SYM7-THEOREM-6.2 and SYM7-TABLE-20-RESULT-21 certify the same Weinberg conclusion $(\Delta B, \Delta L) = (0, 2)$. Their source-authority relation makes the corresponding branch available in SUBGEN-DIM-D5; the independently computed π^{class} of SYM-07-WEINBERG-v1 then equals that branch signature and only then induces its β assignment.

The line-operator sector provides a decisive causal-order test. In the frozen D-SM-G6 line/surface-operator space, several branches are source-available simultaneously. The independent class-side projection gives:

$$\begin{aligned} \text{SYM-09-WILSON-G6-v1} &\mapsto \text{electric line-category interface,} \\ \text{SYM-09-EW-LINES-G6-v1} &\mapsto \text{electroweak Higgs-phase line interface,} \\ \text{SYM-09-MAG-G6-v1} &\mapsto \text{magnetic line-lattice interface.} \end{aligned}$$

The line-polarization record is exact at SUBGEN-BND-LINE-POLARIZATION; its nonboundary occurrences receive explicit different-locus dispositions. The electric Wilson or electroweak line records cannot be assigned to the magnetic line-lattice branch merely because that branch is source-available in the same NF space.

8.4.2 Source-clause range theorem and the falsification branch

For the complete Q06 source-clause partition, define

$$\begin{aligned} N_{\text{occ},\perp} &= \{[x] \in \mathcal{N} : \mathfrak{B}^{\text{cap}}([x]) \text{ and } \mathfrak{B}^{\text{up}}([x]) \text{ both fail}\}, \\ N_{\Delta} &= \{[x] \in \mathcal{N} : \mathfrak{B}^{\text{cap}}([x]) \text{ or } \mathfrak{B}^{\text{up}}([x]) \text{ holds, but} \\ &\quad \pi^{\text{class}}([x]) \notin \pi^{\text{branch}}(\mathcal{C}_{D,\ell,\alpha}^{\text{src}})\}. \end{aligned}$$

Then

$$\mathcal{N} = \left(\bigsqcup_{c \in \mathcal{C}_{D,\ell,\alpha}^{\text{src}}} F_{\mathcal{N},c} \right) \sqcup N_{\text{occ},\perp} \sqcup N_{\Delta}.$$

The branch $N_{\text{occ},\perp}$ may be nonempty. It supplies no occupancy edge under the frozen capstone or imported predicates, but no global rolelessness, physical absence, or materialized nonliveness follows. A member of N_{Δ} is instead a same-target occupancy-positive physical signature outside the source-derived Q06 branch range. It is a faithful counterexample and reopens Q06/Q10.

Lemma 8.5 (Q06 class-projection range and source-branch totality). *For every frozen target/subgenerator pair, the Q06 source-clause partition exhausts the possible occupancy-relevant values of π^{class} . Equivalently, at the frozen source packet $N_{\Delta} = \emptyset$.*

Proof. Fix (D, ℓ, α) and an occupancy-relevant class $[x]$. By Q06, the class is represented by one of the exhaustive source-normal-form alternatives in that exact target/subgenerator domain, with every coordinate entering π^{class} retained. Projecting the complete source-clause disjunction through those retained coordinates yields the branch-signature range $\pi^{\text{branch}}(\mathcal{C}_{D,\ell,\alpha}^{\text{src}})$. Source restatements are quotiented only by certified theorem/result or theorem/theorem derivations in Src; no role or residue population enters this projection. Therefore the independently evaluated tuple $\pi^{\text{class}}([x])$ equals one source-available branch tuple. A same-target tuple absent from this range would be a new Q06 source alternative and hence a counterexample to the frozen Q06 totality theorem rather than something that may be relabeled into $N_{\text{occ},\perp}$. $\square$

The finite named-class audit is downstream implementation evidence, not the semantic totality premise. All 341 named target-specialized Q03 occurrences have one explicit disposition: 270 reconstruct

a unique β from independently computed class-signature equality, 45 have a proved different-Q06-locus disposition, and 26 are composite projections whose occupancy-bearing constituents are resolved at certified loci. None is silently assigned a branch by source co-occurrence.

8.5 Downstream realization and exact compatibility transport

Only after π^{class} , the Q06 branch range, branch-to-space source availability, and the equality-induced β map are fixed are capstone roles and materialized residues read. Applying each independently formed capstone burden to the capstone-local source branches gives

$$\mu_R : R_D^{\text{cap}} \longrightarrow \mathcal{C}_D^{\text{src, cap}}.$$

The observed 130-to-130 correspondence is therefore a theorem output, not a premise for class projection or branch completeness.

Separately, every materialized physical residue is normalized by the same physical tuple, giving

$$\mu_X : \mathcal{C}_D^{\text{src, cap}} \longrightarrow X_D^{\text{mat}},$$

with 130 exact realization certificates. Pointwise E , assignment-level J , liveness, and the materialized singleton-orbit theorem remain later tests.

For every capstone-local exact specialized fiber define

$$P([x]) = \mu_X(\beta([x])).$$

Because β is induced by exact physical-signature equality, P preserves every field inspected by J . Hence, for the same frozen compatibility contract,

$$J_{D,p}^{\text{spec}}(\sigma) \iff J_{D,p}^{\text{mat}}(P \circ \sigma).$$

The construction contains 208 pointwise transport certificates and 42 complete assignment-level biconditionals.

For imported SYM-1–SYM-5 source branches, the same equality-induced factorization defines

$$P_{\text{up}} : F_{N,c} \longrightarrow \Sigma_{D,p}^{\text{up}} / \Gamma_{D,p}^{\text{up}},$$

and the 117 transport certificates land in the already accepted upstream singleton occupancy orbits.

The D-SM-G6 electric/magnetic comparison keeps pointwise eligibility and assignment compatibility separate. Broad pointwise E permits an electric/magnetic one-form swap, but the exact physical interfaces differ. The swapped assignment fails transported J , while the correctly interfaced class-first fibers map to the unique materialized lawful orbit.

Theorem 8.6 (Independent class-first full-domain Q10 lift). *For every frozen target, each target-specialized Q03 class receives its physical signature from π^{class} before any branch is selected. Branch-to-space source relations establish availability only. The memberwise map β is induced solely by exact equality with a source-available Q06 branch signature. The specialized domain therefore decomposes into 325 certified source-branch incidences, a possibly nonempty $N_{\text{occ},\perp}$, and an empty N_Δ . Capstone fibers factor through P and exact assignment-level J transport; imported SYM-1–SYM-5 fibers factor through P_{up} . Adjoining the target-specialized Q03 class spaces creates no second lawful role-assignment orbit and omits no occupancy-relevant Q06 source conclusion.*

Proof. The Q06 class-projection range theorem first constructs the physical range independently of roles and residues. The class-side and branch-side tuples share one extensional schema but are computed independently. Exact equality in a source-available Q06 space then induces β ; source co-occurrence cannot do so. The range theorem proves $N_\Delta = \emptyset$, while $N_{\text{occ},\perp}$ is retained without a global absence or nonlive claim. Only afterward do the independently formed role burdens and materialized residues

realize the capstone branches. Materialized Q10 supplies the unique lawful orbit, and preservation of every J coordinate proves biconditional transport through P ; imported early-sector branches transport through P_{up} . A second specialized lawful orbit would therefore induce a second accepted downstream orbit, fail an exact compatibility predicate, or contain a novel class-side physical signature in N_{Δ} , contradicting a preceding theorem. $\square$

8.6 Blind cross-sector projections

For each of sixteen overlap interfaces I , two separate extractors construct

$$\pi_p(I), \quad \pi_q(I)$$

from the two source theorem sets. Neither extractor can read the other side, the compatibility contract, the atlas, or an expected result. A third object supplies only the typed relation

$$C_I(\pi_p(I), \pi_q(I)).$$

The 32 blind projections and 16 equalizers all satisfy their relations. A one-sided physical mutation changes that projection and fails the equalizer; a downstream atlas mutation changes neither projection.

Theorem 8.7 (Live-residue and unique compatible occupancy, SYM11-Q09/Q10). *For every closure-eligible target D , the source-first local assignment spaces, the Q06-source-branch lift on the full domain $X_D = A_D/\equiv_D$, and the blind cross-sector compatibility relations form a global fiber product*

$$\Sigma_D^{\text{glob}} = \left(\prod_p \Sigma_{D,p} \right) \times_{\mathcal{I}_D} \text{Compat}(\mathcal{I}_D),$$

and

$$\left| \Sigma_D^{\text{glob}} / \Gamma_D^{\text{lawful}} \right| = 1.$$

On the materialized domain, pointwise eligible residues absent from every compatible complete assignment are not live occupants. On the specialized domain, every occupancy-relevant exact source branch is transported by P or P_{up} ; $N_{\text{occ},\perp}$ contributes no frozen occupancy edge and receives no global roleless, absence, or nonlive label.

Proof. The role and residue formation theorems establish independent materialized R and X . Complete pair evaluation establishes E , and enumeration followed by the independently frozen physical contracts establishes materialized J and the local singleton orbits. Independently and earlier, the Q06 source clauses derive the full occupancy-relevant physical branch basis. Source-branch exhaustion eliminates N_{Δ} ; exact shared signatures define P and P_{up} ; capstone assignment compatibility transports biconditionally through P ; and imported early-sector branches land in their accepted singleton orbits through P_{up} . The blind projection theorems establish the overlap values independently and the typed contracts establish all nontrivial equalizers. A fiber product of the resulting singleton local lawful orbits over compatible interfaces has one lawful global orbit. A second inequivalent global assignment would induce a second local orbit, violate exact transport, violate an overlap predicate, or expose a new Q06 source branch, contradicting the corresponding source-first theorem. $\square$

9 Source-licensed carrier support and relative removal minimality

Support is proved after candidate classification and role occupancy. The required physical interface is formed without naming a provider. Carriers are reconstructed from their upstream definitions, and direct support eligibility is derived from explicit source licenses for the operations required by the neutral interface. Carrier sort is descriptive metadata only: sort equality or inequality is never a premise of provider eligibility.

9.1 Carrier-neutral support requirements

Every support-bearing theorem obligation is routed to a neutral interface type. The rule depends on the target, record locus, physical quotient, anomaly, vacuum, boundary, operator, scale, or witness requirement, not on a carrier identity. The resulting 16167 requirements cover every canonical atlas record. A support-bearing dependency outside this vocabulary reopens Q11.

9.2 Source-licensed direct operations

For each carrier K , let $\text{Lic}(K)$ be the finite set of direct operations explicitly licensed by the cited upstream source facts. A license states that the carrier constructs, witnesses, transports, constrains, fixes, or otherwise supplies that operation at the frozen interface. It is a theorem about the declared direct support graph; it is not a theorem that other carriers are physically incapable of related support through a different direct theorem or a composite path.

For a neutral capability p with required operation set $\text{Req}(p)$ define the direct-provider predicate

$$\text{Cap}_p^{\text{dir}}(K) \iff \text{Req}(p) \subseteq \text{Lic}(K).$$

The capability predicate is frozen before carrier inspection and contains no expected carrier ID and no required carrier-sort equality.

For every carrier/capability pair the source-licensed decision is:

- **YES:** every required direct operation has an explicit upstream source license for K ;
- **NO-DIRECT-LICENSE:** at least one required direct operation has no published direct license for K in the frozen source-rule corpus;
- **OPEN/REOPENED:** a new direct license or an explicit source-backed cross-carrier morphism/path is introduced and must be evaluated before Q12 can close again.

A NO-DIRECT-LICENSE result is therefore relative to the declared direct graph. It does not assert physical impossibility and is not inferred from a carrier tag or from absence in a synthetic property profile.

Define the direct provider set

$$\text{Prov}^{\text{dir}}(p) = \{K : \text{Cap}_p^{\text{dir}}(K) = \text{YES}\}.$$

Multiple direct providers are permitted. A newly licensed second provider expands this set and reopens the removal calculation rather than being excluded by definition.

Theorem 9.1 (Support completeness, SYM11-Q11). *Every structural, bridge/fixation, quotient, obstruction, operator, boundary, continuation, or witness interface required by the complete closure-eligible scientific atlas, materialized or target-specialized, has at least one source-licensed direct provider or an explicitly source-backed support path.*

Proof. The support-dependency map covers every support-bearing theorem obligation. Each neutral interface supplies its required operation set independently of carrier identity. The source-license relation is then evaluated carrier by carrier. Every one of the 16167 interfaces has a nonempty direct provider set in the frozen graph. No expected provider, carrier-sort match, atlas role, or prior support edge enters the decision. An additional source-backed morphism/path would be represented explicitly and would reopen the affected removal proof. $\square$

9.3 Actual deletion and provider recomputation

For a retained direct edge $e = (K, p)$, recompute

$$\text{Prov}_{-e}^{\text{dir}}(p) = \text{Prov}^{\text{dir}}(p) \setminus \{K\}.$$

If a second licensed direct provider survives, the edge is redundant in the declared direct graph unless a separately proved distinction shows that the two edges realize different neutral interfaces. If an explicit source-backed composite route is introduced, the same interface must be reevaluated before the edge can remain in a minimal support presentation.

Theorem 9.2 (Removal-minimal support, SYM11-Q12). *Every published support edge passes the frozen removal test relative to the declared source-licensed direct support graph. Across all 16167 retained edges, deletion leaves no alternate source-licensed direct provider for the same neutral interface; consequently the interface loses its declared direct witness/transport/fixation path and the edge is indispensable in this published support presentation.*

Proof. The neutral requirement is fixed before provider inspection. Provider sets are computed solely from the upstream direct-operation licenses. Each retained edge supplies one of those required interfaces. Removing the edge and recomputing the direct provider set yields no alternate licensed direct provider for the same interface. Therefore the published direct support presentation loses the required direct witness or transport path. This is precisely the skeleton's frozen relative-minimality claim $\chi_D = \text{Min}_{\text{remove}}(\chi_D)$; it is not metaphysical uniqueness of all possible explanatory graphs. A new source-backed license or morphism reopens the theorem rather than being ruled out by ontology. $\square$

What would break Q11/Q12

A support-bearing theorem obligation outside the neutral interface grammar breaks Q11. A new upstream direct-operation license that makes a second carrier satisfy the same capability, or an explicit source-backed alternate support path, reopens Q12 and forces provider/removal recomputation. Carrier-sort inequality alone is never a valid exclusion. A support edge is not protected merely because the published table contains one provider.

The complete neutral-interface grammar, direct-operation licenses, provider recomputation, and full-atlas support lift are given in Sec. 9 and Apps. E and K.6 of the audit-grade theorem volume [1, Sec. 9 and Apps. E, K.6] and in Sec. 2.6 of the source-first report [3, Sec. 2.6].

IV

Integration, rivals, and exhaustion

10 Cross-paper integration and overlap reconciliation

The capstone is not a concatenation of eleven tables. Rows shared by several papers must have one target, one status vector, one extensional identity, and compatible role/support assignments.

10.1 Conflict-resolution order

When two sector exports appear to disagree, the resolution order is:

target $\rightarrow$ acting and acted-upon objects $\rightarrow$ physical quotient $\rightarrow$ quantum/anomaly,
 $\rightarrow$ vacuum, boundary, operator, and scale status $\rightarrow$ role $\rightarrow$ presentation.

A later role label cannot repair a mismatch in an earlier physical field. A genuine target difference becomes a target-update edge; an alias or basis difference is normalized only after the physical fields agree.

10.2 Mandatory overlap cases

SYM-0 through SYM-3 foundation-source disposition. The common status calculus is received as calibration ancestry, not duplicated as physical population. Its forty-six benchmark/status records map to the downstream target-qualified objects whose QFT status they calibrate. The fifty-seven SYM-1 gauge-foundation records and thirteen SYM-2 spacetime facets are dispositioned through explicit source crosswalks into target-qualified gauge/spacetime classes or theorem-level interfaces. SYM-3's ten anomaly-interface records are likewise reconciled separately; the $SU(2)$ Witten global anomaly is explicit rather than being absorbed into the perturbative anomaly row. This foundation crosswalk prevents a coarse owner label from masking a missing anomaly, boundary, or active/passive action distinction.

SYM-4 discrete-source reconciliation. The seventy-nine frozen atomic discrete rows are received at their source target, not collapsed into one theory-wide $C/P/T/CP$ bit. One-generation, two-generation, and diagnostic control targets remain outside the closure family; generic and special strong-CP targets stay distinct; CKM CP-zero and degeneracy controls remain target-qualified; and the integrated $C/P/T$ compiler rows are cross-target summaries rather than fixed-fiber occupants. Thus a target-qualified discrete action can be absorbed only by an extensional-equality proof at that same target.

SYM-5 flavor-source reconciliation. The eighty-two frozen flavor rows are likewise dispositioned at source granularity. The massless-neutrino, zero-Yukawa, generic-quark, Dirac-neutrino, Weinberg/Majorana-EFT, higher-flavor, MFV-control, and flavor-strata fibers remain separate. Generic-quark orbit/invariant/chart data and Majorana/Weinberg κ -tensor data are explicit where required, while MFV and stratification controls remain outside closure unless separately admitted. This prevents basis covariance or a mixing-chart label from standing in for the invariant flavor population.

Gauge architecture, local operation, and boundary action. SYM-1 separates the load-bearing gauge package from proper compact-support operations and from boundary/large operations. SYM-3 imports the gauge-anomaly consistency field. SYM-8 imports the quotient split into the electroweak fixation target, while SYM-9 refines the global-form and extended-operator target. The capstone therefore has distinct rows rather than one "gauge symmetry" bit.

CPT, CP, and flavor basis. SYM-2 supplies the CPT premise package. SYM-4 uses it to type sector-local CP/T relations and physical CP invariants. SYM-5 supplies the basis quotient and fixed-tensor stabilizers. The canonical CP row is therefore indexed by target and invariant data; a complex matrix representative does not create CP violation without the invariant certificate.

Generation cardinality and flavor target. The core symmetry target receives $N_g = 3$ from the independent Carrier-ARC role-occupancy endpoint rather than from CKM/PMNS matrix size or atlas population. SYM-5 and SYM-7 therefore classify flavor, mixing, family differences, and remnant structure relative to the fixed three-generation Standard Model target, while an additional-generation proposal is a target update unless it defeats the upstream cardinality theorem.

Fermion parity and faithful remnant quotients. SYM-7 uses the order-two fermion-parity action in faithful quotienting of baryon/lepton remnants. SYM-11 therefore publishes an explicit fermion-parity/Spin-center overlap record rather than leaving $(-1)^F$ as an untyped ingredient. The record is a quotient/center-overlap datum, not a newly promoted primitive internal symmetry.

Baryon/lepton actions and anomaly. SYM-7 owns operator-dimension and neutrino-target status. SYM-3 supplies the electroweak anomaly/topology field. SYM-9 supplies any extended/global-form

refinement. The atlas distinguishes perturbative accidental action, nonperturbative violation, SMEFT breaking, and finite remnants.

SYM-6 atomic projection and vector-baryon action. The SYM-6 source inventory is received at atomic rather than family-label granularity. In the chiral QCD fiber, the faithful nonsinglet action, vector $U(1)_B$ action/baryon normalization, vacuum/nonlinear realization, continuous and discrete axial statuses, and background-anomaly matching are separately typed. The exact and physical isospin and hadronic- $SU(3)$ records occupy different target fibers; custodial matter branches are not folded into the gauge-Higgs branch; and the four HQET atomic records are separated by heavy-species and finite-order matching scope. The complete eighteen-record crosswalk is supplied in the audit-grade theorem volume. This source-to-population proof ensures that the CHI-VECTOR record is not omitted.

Custodial and electromagnetic structure. SYM-6 owns custodial target fibers; SYM-8 owns the electromagnetic stabilizer and physical charge/spectrum action. The two are not merged: $SU(2)_V$ and $U(1)_{em}$ act on different objects under different target assumptions.

Scale, RG, and effective symmetry. SYM-6 supplies target-indexed IR limits and matching obligations; SYM-10 supplies quotient RG transport, fixed-point status, EFT-envelope matching, and emergence provenance. A scale change is not automatically a physical dilatation action, and an IR symmetry is not automatically a UV anchor.

Global form, one-form action, and theta/phase. SYM-9 distinguishes local algebra, quotient group, line polarization, screening, monopole policy, gauge-topological theta torus, and full physical phase quotient. Discrete CP rows from SYM-4 are imported only after those target coordinates are fixed.

SYM-8 final-publication projection. The sixteen-record SYM-8 publication atlas is reconciled against nine source target schemas. The one-doublet electroweak fiber carries the frozen perturbative spectral bundle where the physical-spectrum theorem is invoked. The algebraic response rank/kernel record is therefore not used as a surrogate for the independent BRST/pole/invariant-correlator spectrum record. Neutral A/Z normal form, electromagnetic generator and point-sector action, proper-local electromagnetic quotient, boundary/asymptotic action, Goldstone translation, custodial noncollapse, and the integrated presentation normal form are separately dispositioned whenever their extensional signatures differ.

SYM-9 final-publication projection. The twenty-one-record global-form source atlas is reconciled before family closure. Electric Wilson descent, matter screening, residual electric one-form action, magnetic/dyonic line structure, magnetic one-form action, electroweak line transport, theta identification, and physical phase quotient are not collapsed merely because they belong to one global-completion packet. Fixed- Γ target identity, line branch, monopole policy, and background/fractionalization data remain explicit.

SYM-10 final-publication projection. The nineteen-record scale/RG source atlas is likewise reconciled at source granularity. Passive scale redescription, quotient RG transport, fixed-point status, EFT-envelope identity, threshold matching, intrinsic IR symmetry, UV-to-IR emergence ancestry, dimensional transmutation, and the restricted V_0 correspondence retain their distinct objects and target conditions. In particular, physical IR symmetry is not inferred from an emergence claim, and an emergence claim is not used as a surrogate for the intrinsic IR action.

The complete ownership map is supplied in the technical supplement.

Proof spine

The overlap reconciliation establishes the joint-interface premise used by SYM11-Q10. Every shared row has one authoritative owner for each field and one frozen import path for cross-cutting anomaly, quotient, target, and scale data. No field is resolved by choosing whichever paper gives the preferred endpoint.

10.3 SYM-7 target-qualified accidental-symmetry projection

SYM-7 is reconciled at source granularity rather than by a paper-level ownership label. The source crosswalk covers all thirty-four canonical results and all thirteen registered source target schemas. The electroweak-topology target therefore contains separate target-qualified records for baryon transfer, family-lepton transfer, the $B + L$ violated direction, the $B - L$ null direction, the two family-difference null directions, the topological zero-mode vertex, and the continuous null-lattice remnant. The neutrino branch keeps the renormalizable three- ν_R Majorana target distinct from the low-energy Weinberg EFT. The $d = 6$ ν SMEFT, corrected $d = 7$ LNV and BLNV, selected $d = 9$ $\Delta B = 2$, and two $B - L$ anomaly-background targets receive their own target-qualified records. Finally, the twelve frozen remnant effect sets are represented at their actual component fibers rather than compressed into one generic remnant label. This source-to-capstone reconciliation discharges the SYM-7 portion of SYM11-Q06; the crosswalk file is an implementation surface for that scientific mapping, not its theorem authority.

11 Construction coherence, versioning, and supersession

An atlas can be internally consistent at one snapshot and still be illegitimate if its rows were repaired after a failure by changing their target, source, or meaning. The construction history must therefore preserve ancestry.

Definition 11.1 (Atlas construction trajectory). A construction trajectory is a finite sequence

$$\tau = (\mathfrak{A}^{(0)}, u_1, \mathfrak{A}^{(1)}, \dots, u_n, \mathfrak{A}^{(n)})$$

where each u_i is a typed operation: add, split, merge, alias, target-update, source correction, supersession, or presentation edit. It is coherent when every operation is lawful at the state at which it occurs and every preserved record keeps its extensional identity and source ancestry.

Theorem 11.2 (Construction coherence and no deferred legitimacy, SYM11-Q13). *The canonical atlas has a coherent construction trajectory. Every current row has an unbroken ancestry to its owning sector theorem or is explicitly introduced as a new capstone normalization row; every superseded row remains traceable; every split, merge, and target update is typed at the stage at which it occurs; and no invalid row is made legitimate solely by a later target change or source replacement.*

Proof. The trajectory rules are the specialization of prefix and composition coherence from [38]. Each current row records its owner, source certificate, introduction/change history, and stable theorem IDs. A split preserves the parent pointer while assigning distinct extensional signatures. A merge requires prior proof of lawful equivalence. A target update uses a registered arrow and never retroactively converts a failed same-target candidate into a success. A source correction supersedes the old claim rather than silently altering it. Because every operation is checked at its application stage and composition preserves the same target/signature relations, the full trajectory is coherent. The supersession graph is acyclic, so no row derives its authority from a descendant. $\square$

Certificate

The version and supersession registries are publication evidence, not the proof of the sector physics. They certify that the statement proved and the statement published have the same identity and ancestry.

12 Disjoint rival disposition and the genuine-counterexample branch

A closure theorem is credible only if its counterexample branch remains open until the scientific argument has eliminated it. The rival protocol begins with the proposal, not with a desired disposition. Lawful skin changes are already members of the redescription equivalence class and therefore are not given a second overlapping disposition.

Definition 12.1 (Disjoint rival classes). Let s' be a proposed symmetry-relevant record compared with the canonical atlas at target D .

- I. **Absorbed/equivalent:** s' lies in an existing $\equiv_D$ class, including lawful gauge, basis, normalization, scheme, coordinate, or other skin redescription.
- II. **Same-target obstructed:** the complete target agrees, but s' fails a required action, quotient, anomaly, vacuum, operator, compatibility, or scale gate and therefore does not define an admissible new atlas class.
- III. **Target update:** at least one target-bearing coordinate differs.
- IV. **Genuine same-target counterexample:** the complete target agrees, all mandatory physical/status and compatibility gates pass, s' is not equivalent to an existing class, and it changes same-scope physical or standing-bearing content.

Theorem 12.2 (Disjoint rival disposition theorem, SYM11-Q14). *For every symmetry-relevant proposal expressible in the frozen record language, exactly one of the four rival classes holds.*

Proof. First compare target fingerprints. A difference gives class III and excludes the same-target classes. At fixed target, normalize all lawful presentations. If the normalized extensional signature is already present, class I holds. Otherwise test the mandatory target-qualified QFT/status and joint-compatibility gates. Failure of any such gate gives class II. If all gates pass while the class remains new and changes physical/standing content, class IV holds. The tests are ordered and mutually exclusive, and SYM11-Q05–SYM11-Q06 ensure that every expressible proposal reaches one of them. $\square$

12.1 Counterexample burden

A class-IV proposal must state the exact target, acting and acted-upon objects, action, quotient, invariant, anomaly, vacuum, boundary/global-form, operator, and scale status; identify the new role or support claim; show non-equivalence to existing rows; and name the theorem it contradicts. It is not required to conform to the existing atlas if its point is to falsify the schema or locus theorem.

Adversarial check

The capstone may not move a class-IV proposal into class III after the same-target proof fails unless an independently changed target coordinate is exhibited. Nor may it call a new physical action “skin” merely because its group name is familiar. Conversely, lawful skin is absorbed into class I and is never double-counted as a separate rival category.

The companion rival ledger contains representative resolved rivals and an explicit protocol row for a future genuine counterexample. Its genuine-counterexample data file is an implementation reflection only; its emptiness is not a premise of the theorem.

13 Source-first genuine-rival elimination

Q14 leaves the genuine same-target branch open. It is closed only after Q05 formation, Q06 total scientific disposition, independent Q10 occupancy, Q11/Q12 support, and finite composite closure have all been proved.

13.1 Local candidate disposition

Let x be a generated same-target proposal. Q05 assigns it to a unique locus and selector subfamily. The corresponding source-first bridge gives exactly one branch:

$$\text{NF}, \quad \text{OBS}, \quad \text{TGT}, \quad \text{OPEN}.$$

At a closure-eligible target, OPEN is absent because every Q05-admitted target/subfamily pair remains an explicit Q06 burden and every exact source-domain inclusion, source theorem application, remainder classification, and semantic NF pullback theorem is present. TGT is available only when the candidate changes an actual frozen target coordinate; absence of an NF representative does not create TGT or remove the pair from Q06. OBS does not survive the required same-target equations. NF belongs to the target-qualified source normal-form class space $\mathcal{N}_{D,\ell,\alpha} \subseteq \mathfrak{A}_D/\equiv_D$; distinct lawful NF classes are retained rather than collapsed.

13.2 Composite closure

For a finite same-target composite $x_1 \star \cdots \star x_n$, order the physical differences by the Q05 locus order. Apply the local classifier at the first difference and evaluate every later component through the complete pairwise interface calculus. A failed dependent interface is OBS; imported target-bearing structure is TGT; missing theorem authority is OPEN; otherwise the compatible NF components form an already typed product/equalizer orbit. Induction on n closes every finite composite without brute-force enumeration.

Theorem 13.1 (Genuine-rival elimination, SYM11-Q15). *For every $D \in \mathfrak{D}_{\text{SM}}^{\text{cl}}$, the generated class-IV same-target rival branch is empty.*

Proof. Assume a generated class-IV rival x . Q05 places it in one unique subfamily or finite composite. Q06 and composite closure give NF/OBS/TGT/OPEN. TGT contradicts equality of target; OBS contradicts survival; OPEN is absent by the proved source-first target/subfamily cells; and NF contradicts x being outside the canonical target-qualified normal-form class space. Therefore no generated class-IV rival survives. $\square$

What would reopen Q15

Q15 fails if a new same-target first difference escapes Q05, a Q05-admitted target is removed from Q06 because its NF map is empty, TGT is asserted without a real target-coordinate change, a bridge loses an identity-bearing coordinate, a source-satisfying same-target candidate is omitted from its target-specialized NF class space, an exact source theorem does not classify the projected branch, a source-domain remainder is unclassified, a second NF orbit exists, a composite escapes the pairwise interface calculus, or any closure-target cell becomes OPEN. An empty rival ledger cannot repair any of these failures.

14 Atlas assembly, fiberwise uniqueness, and same-scope exhaustion

Fix $D \in \mathfrak{D}_{\text{SM}}^{\text{cl}}$. Q01–Q04 provide the well-typed extensional quotient $\mathfrak{A}_D/\equiv_D$. Q05–Q06 provide independent candidate formation and total disposition. Q07–Q10 provide authority factorization and one compatible role-occupancy orbit. Q11–Q12 provide complete removal-minimal support. Q13 preserves dependency history, and Q14–Q15 give exact rival disposition and class-IV emptiness.

Theorem 14.1 (Canonical atlas assembly, SYM11-A01). *These theorem layers assemble a canonical target-qualified atlas quotient $\mathfrak{A}_D/\equiv_D$. Its scientific normal-form object is the union of the materialized typed-record classes and the theorem-defined target-specialized class spaces $N_{D,\ell,\alpha}$; named rows are a finite publication view, not a premise for inherited NF existence. Every materialized class and every target-specialized NF class/subspace lies in the final role/support theorem domain before assembly. Materialized rows carry complete physical/status data. Specialized classes carry their source-normal-form identity, provenance-independent physical signature, separate source authority, certified exact branch transport or explicit different-locus/composite source disposition, lifted support ancestry, and rival disposition. The occupancy-neutral complement carries no global roleless, physical-absence, or nonlive label.*

Proof. Q01 supplies complete typed rows. Q02 separates same-target comparison from update arrows. Q03/Q04 quotient only identity-preserving redescription. Q06 fixes every generated physical/status packet before role assignment. Q10 supplies the unique compatible role orbit on the full $X_D = A_D/\equiv_D$: theorem/result source-conclusion equivalence first derives the exact specialized branch incidences; capstone fibers then transport through P/J , imported early-sector fibers through P_{up} , and every known nonfactorizing named occurrence has an explicit different-locus or composite source theorem rather than a silent zero incidence. Q11/Q12 attach the complete relative-minimal support graph after the specialized source-classifier support lift. Q13 carries provenance and supersession without deferred repair. Q15 removes the generated class-IV branch. $\square$

Theorem 14.2 (Fiberwise atlas uniqueness, SYM11-A02). *Let $\mathfrak{A}'_D$ be any complete admissible atlas satisfying the same frozen target, record language, candidate protocol, source-theorem hypotheses, lawful equivalence, role architecture and support criteria. Then*

$$\mathfrak{A}'_D/\equiv_D \cong \mathfrak{A}_D/\equiv_D.$$

The isomorphism preserves target, extensional physical/status identity, role incidence, support ancestry and source/evidence status, but need not preserve row order, names, notation or other skin.

Proof. Take any class $[x] \in \mathfrak{A}'_D/\equiv_D$. Q05 places its first physical difference in a unique formation locus and Q06 in a unique subfamily disposition. OBS and TGT are incompatible with membership in a complete same-target admissible atlas; OPEN is absent in $\mathfrak{D}_{\text{SM}}^{\text{cl}}$; hence $[x]_{\equiv_D}$ lies in the corresponding target-specialized NF class space contained in $\mathfrak{A}_D/\equiv_D$. This gives a well-defined map from $\mathfrak{A}'_D/\equiv_D$ to $\mathfrak{A}_D/\equiv_D$. Reversing the argument for the complete canonical atlas gives surjectivity. Q03 gives injectivity on extensional classes. The full-domain Q10 theorem forces every occupancy-relevant specialized branch into the same lawful role orbit through P or P_{up} ; explicit different-locus/composite occurrences create no occupancy edge and receive no nonlive inference, while a new occupancy-positive signature reopens N_Δ . The full-domain Q11/Q12 theorem forces the same required carrier-neutral interfaces and retained relative-minimal providers/support ancestry. Thus the map is the stated lawful isomorphism. $\square$

Theorem 14.3 (Achieved same-scope exhaustion, SYM11-A03). *For every $D \in \mathfrak{D}_{\text{SM}}^{\text{cl}}$*

$$\text{Surv}_D^{\text{sym}}(C_D \setminus \mathfrak{A}_D) = \emptyset.$$

Proof. Assume $x \in \text{Surv}_D^{\text{sym}}(C_D \setminus \mathfrak{A}_D)$. Q14 assigns it to exactly one of equivalence/skin, same-target obstruction, target update, or genuine same-target counterexample. The first branch contradicts $x \notin \mathfrak{A}_D$ modulo $\equiv_D$; the second contradicts survival; the third contradicts fixed target; and Q15 empties the fourth. Hence no such x exists. $\square$

Proof and certification status

A02 and A03 are relative theorems. They quantify over the explicitly frozen target family, record language, candidate constructors and same-scope comparison class. A BSM field, new operator category, different global completion, new background or genuinely new first-difference locus is

not rejected by fiat: it changes target, extends the declared language, or falsifies the corresponding completeness theorem.

The resulting target, role, support, rival, and provenance projections can be inspected row-by-row in the atlas companion [2, Target, role, support, rival, and provenance views]; those views are downstream projections of A01–A03 rather than premises for them.

15 Target-indexed family closure and principal theorem

Theorem 15.1 (Family assembly, SYM11-A04). *The canonical fiber quotients assemble over the licensed target-update graph into*

$$\mathfrak{A}_{\text{SM}} = \{[\mathfrak{A}_D] : D \in \mathfrak{D}_{\text{SM}}^{\text{sym}}\}.$$

Update arrows preserve declared imports and explicitly invalidate claims whose target hypotheses change. Distinct field-content, generation-governance, phase, global-form, line, monopole, boundary, operator, neutrino, background and RG/EFT fibers are not collapsed into one physical theory.

Proof. F01 freezes the target graph, including the Carrier-ARC handoff $N_g = 3$ as target governance rather than a symmetry row or primitive generation carrier. Q02 separates same-target identity from update. Q13 preserves fixed-fiber dependency history and forbids silent backflow. Since $\equiv_D$ never identifies target-bearing differences, the canonical quotients assemble functorially over the licensed update arrows. $\square$

Theorem 15.2 (Target-indexed Standard Model symmetry atlas uniqueness and same-scope exhaustion, SYM11-MAIN). *For the declared Standard Model symmetry target family:*

1. *the target, import and record substrate is frozen and non-circular;*
2. *every admitted same-target physical/status difference enters one of nine independent formation spaces or openly defeats Q05;*
3. *every formed candidate enters exactly one of 54 typed subfamilies, whose exact theorem hypotheses and NF/OBS/TGT/OPEN totality are proved before atlas projection, with target-specialized source NF class spaces on every Q05-admitted fiber;*
4. *the complete 45-pair composite calculus introduces no untyped fifth branch;*
5. *lawful redescription descends to canonical extensional classes;*
6. *fixed-target authority factors through Anchor, Tensor or Skin, refined by the declared functional roles;*
7. *the independent Q06 class-side π^{class} map, independently derived source-branch range, branch-to-space source availability with no memberwise authority, exact equality-induced β , source-clause exhaustion of N_Δ , complete named exact/different-locus/composite disposition, independent local R/X constructions, complete materialized E/J, exact capstone P/J transport, imported SYM-1–SYM-5 P_{up} transport, and source-projection equalizers yield one global lawful role-occupancy orbit on the full $A_D/\equiv_D$ domain;*
8. *every materialized or specialized carrier-neutral support interface has source ancestry and a provider, and every retained edge is indispensable under provider recomputation from carrier-neutral required operations and explicit upstream direct-operation licenses;*
9. *every rival is equivalent/skin, same-target obstructed, target-changing, or a genuine same-target counterexample;*
10. *no generated genuine same-target counterexample survives;*
11. *consequently every closure-eligible fiber is unique up to lawful atlas isomorphism and satisfies same-scope exhaustion; and*
12. *the fiber quotients assemble into the target-indexed atlas normal-form family.*

Therefore, for every $D \in \mathfrak{D}_{\text{SM}}^{\text{cl}}$ and every complete admissible same-target atlas $\mathfrak{A}'_D$,

$$\text{Surv}_D^{\text{sym}}(C_D \setminus \mathfrak{A}_D) = \emptyset, \quad \mathfrak{A}'_D / \equiv_D \cong \mathfrak{A}_D / \equiv_D .$$

The theorem is relative to the declared target family and comparison language; it does not assert uniqueness of the Standard Model under arbitrary premises.

Proof. F01–F03 supply target, import and schema authority. Q01–Q04 supply the extensional quotient. Q05 and the selector/applicability/totality theorems supply candidate and status completeness. Q07–Q10 supply authority and independently derived full-atlas role occupancy. Before the role or residue registries are read, every target-specialized Q03 class receives $\pi^{\text{class}}([x]) = (D, \kappa, F, I, S, V)$ directly from Q06-retained physical coordinates. Independently, the source clauses derive the Q06 branch-signature range and branch-to-space availability relations. Those relations cannot select a memberwise branch. Only exact $\pi^{\text{class}}([x]) = \pi^{\text{branch}}(c)$ in the same source-available Q06 space induces $\beta([x]) = c$. The 54 source-clause range/exhaustion theorems prove $N_{\Delta} = \emptyset$ while leaving $N_{\text{occ}, \perp}$ without a global roleless or nonlive interpretation. The finite named audit checks the implementation of the construction, including the Weinberg $d = 5$ theorem/result relation and the distinct Wilson/electroweak-line/magnetic/line-polarization factorizations, but does not supply semantic totality. Only afterward do capstone burden/branch and branch/residue realization, materialized E/J , exact P/J transport, and imported P_{up} transport operate. Q11–Q13 supply full-atlas carrier-neutral support closure, including the specialized source-classifier support lift and coherent construction. Q14–Q15 give the exact rival partition and scientific class-IV elimination. A01–A03 give fiber assembly, uniqueness and exhaustion; A04 gives family closure. F02 ensures no terminal capstone conclusion is imported as a premise. $\square$

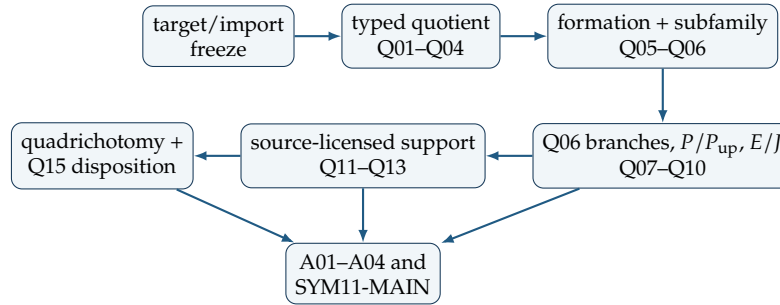


Figure 3: Proof order. Q05/Q06 first derive the target-specialized class spaces. The memberwise class-side π^{class} and the source-branch signature range are then constructed independently. Branch-to-space source relations establish availability only; exact signature equality induces β afterward. Capstone role/residue realization, materialized E/J , P/J transport, imported P_{up} transport, blind sector equalizers, and source-licensed direct support follow downstream. The atlas, support graph, ledgers, and implementation checks are projections rather than premises.

16 Interpretation and exact nonclaims

The capstone result is strong but relative. It says that, once the Standard Model symmetry targets and comparison language are explicit, there is one redescription-stable atlas normal form in each closed fiber and no surviving same-scope symmetry-relevant record outside it. It does not say that the Standard Model is the only possible physical theory.

16.1 What the result establishes

The atlas makes several conceptual points precise.

First, symmetry is not one ontological kind. Gauge architecture, representative operations, spacetime actions, anomaly obstructions, vacuum stabilizers, effective-limit actions, generalized symmetries, and RG transport occupy different statuses and roles.

Second, exactness, physicality, and structural priority remain independent. CPT can be exact and theorem-derived; an approximate chiral or heavy-quark symmetry can act physically in its target; a gauge operation can be skin while gauge architecture is load-bearing.

Third, target indexing is part of physical identity. A global-form completion, defect category, neutrino sector, operator truncation, massless limit, fixed point, or EFT envelope cannot be imported into the sharp-local core without an update arrow.

Fourth, lawful redescription preserves rather than erases physics. The atlas quotient removes gauge, basis, normalization, scheme, and notation multiplicity while retaining charges, anomalies, invariants, spectra, defects, matching data, and physical actions.

Fifth, closure requires more than row coverage. It requires a complete candidate-locus theorem, target-qualified population established from the frozen source dispositions, compatible role occupancy, carrier-support closure, and an open rival branch discharged by the science. A source crosswalk is therefore evidence that every frozen sector output has received a scientific disposition; its row count is never itself the completeness proof.

16.2 Exact nonclaims

Even a successful capstone does not establish:

- uniqueness of the Standard Model under arbitrary physical premises;
- a unique physically realized global form Γ ;
- numerical gauge couplings, masses, Yukawa eigenvalues, mixing angles, vacuum scales, or RG integration constants;
- completion over supersymmetry, grand unification, Peccei–Quinn sectors, BSM family symmetries, quantum gravity, or arbitrary UV completions;
- every generalized or non-invertible symmetry of every extension;
- a universal one-to-one correspondence between carriers and symmetries;
- that every nonprimitive structure is unreal, dispensable, or mere notation;
- a machine proof of the interacting Standard Model;
- replacement of dynamics, experiment, or domain expertise by an atlas.

Exact nonclaim boundary

The comparison class is the declared target family and frozen record language. A new physical discovery may add a target node, revise a sector theorem, or falsify candidate completeness. The correct response would be a versioned update, not relabeling designed to protect the present endpoint.

17 Companion artifacts and reproducibility boundary

The publication package separates four reading purposes.

1. **Publication manuscript.** The present paper gives the target family, definitions, theorem statements, proof chain, exact nonclaims, and principal result at journal granularity.
2. **Audit-grade theorem volume.** `SYM11_Standard_Model_Symmetry_Atlas_and_Exhaustion_Closure.pdf` preserves the complete theorem architecture, generated scientific tables, target/sub-family bridges, and hostile-regression appendices of the science-frozen v2.10.0 project [1, Secs. 1–18 and Apps. B–K].

3. **Atlas companion.** `SYM11_Standard_Model_Symmetry_Atlas_Companion.pdf` presents the canonical typed rows and synchronized target, role, support, rival, and provenance views [2, Target-family, candidate/role, support, rival, and provenance views].
4. **Source-first derivation and machine-readable archive.** `release/SYM11_Source_First_Closure_Derivation_v2.10.0.pdf`, together with the CSV/JSON data, source-fact ledger, build scripts, validators, hostile tests, and manifests, records the derivation and implementation boundary [3, Secs. 1–3, especially Secs. 2.2–2.7].

Scientific proof versus certification

The sector theorems and the deductions in this manuscript establish candidate completeness, normal-form identity, compatible occupancy, support closure, rival elimination, and exhaustion. Machine-readable registries, deterministic builds, hashes, validators, and bounded hostile mutations establish that the released artifacts faithfully encode those declared dependencies and fail openly under the registered perturbations. They do not infer the physics or supply semantic exhaustion by row count.

The science-frozen authority is v2.10.0. The publication-clean layer changes title-page presentation, abstract, reader navigation, copyediting, artifact separation, bibliography presentation, PDF metadata, and cross-references only. It introduces no change to the target family, candidate grammar, theorem hypotheses, normal-form spaces, role/residue construction, compatibility predicates, carrier-support relation, rival theorem, principal conclusion, or exact nonclaims.

18 Conclusion

The Standard Model symmetry problem is not exhausted by a list of group names. Gauge architecture and gauge operations, active and passive spacetime transformations, anomaly obstructions, basis covariance, exact and approximate flavor actions, vacuum fixation, global-form and defect data, and RG/EFT structures differ in their targets, quotients, observables, and physical roles. The atlas therefore classifies target-indexed typed statements and quotients only lawful identity-preserving redescription.

Candidate completeness is established before atlas population. Every admitted same-target first difference enters one of nine formation loci and one of 54 typed source-theorem subfamilies. For every Q05-admitted target/subfamily pair, the exact source hypotheses define a target-qualified normal-form class space $\mathcal{N}_{D,\ell,\alpha}$. A named row is only an optional representative of an already derived class; its absence cannot prove that the class is empty, obstructed, or target-changing.

Role occupancy is likewise constructed in the required direction. The theorem burdens and physical residues are extracted independently. For theorem-defined specialized classes, the class-side physical signature $\pi^{\text{class}}([x])$ is computed from Q06-retained coordinates before a source branch, role, residue, eligibility edge, or lawful orbit is selected. Exact equality with an independently derived source-available branch induces the memberwise factorization. Capstone-local fibers then transport through P with assignment-level J preserved, while early-sector fibers transport through P_{up} to their accepted occupancy orbits. The occupancy-neutral complement is not declared physically absent or globally roleless; a same-target occupancy-positive signature outside the source range would instead enter N_{Δ} and reopen Q06/Q10.

Carrier support begins from neutral required operations and explicit upstream direct-operation licenses. Provider sets are formed only after that relation is fixed, and deletion with provider recomputation proves every retained edge indispensable relative to the declared direct-support presentation. A newly supplied direct license or source-backed alternate morphism would reopen the affected minimality theorem rather than being excluded by definition.

Finally, the rival theorem keeps the counterexample branch scientifically open. Any proposal in the frozen record language is lawfully equivalent, same-target obstructed, target-changing, or a genuine

same-target counterexample. Candidate totality, source-domain classification, and finite composite closure eliminate the generated genuine branch. Hence, for every closure-eligible target D and every complete admissible atlas $\mathfrak{A}'_D$ satisfying the same frozen premises,

$$\mathfrak{A}'_D / \equiv_D \cong \mathfrak{A}_D / \equiv_D, \quad \text{Surv}_D^{\text{sym}}(C_D \setminus \mathfrak{A}_D) = \emptyset.$$

The fiber quotients assemble over the licensed target-update graph without collapsing different field contents, phases, global forms, operator categories, neutrino sectors, backgrounds, or RG/EFT regimes.

The result is intentionally relative. It does not establish uniqueness of the Standard Model under arbitrary physical assumptions, choose one empirically realized global form, determine masses or couplings, close every BSM or quantum-gravity extension, or reduce descriptive structure to unreality. A new target, unrouteable physical field, unclassified source-domain remainder, second normal-form orbit, novel occupancy-positive signature, failed compatibility transport, uncovered support interface, alternate provider defeating removal minimality, or faithful same-target class-IV survivor would reopen the corresponding theorem. That falsification boundary is part of the result.

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